Linear Regression using Python

Online Advertising-Dollars (in 1000s)	Monthly E-commerce Sales-Dollars (in 1000s)
1.7	368
1.5	340
2.8	665
5	954
1.3	331
2.2	556
4	800
3.5	700

What is Regression?

- A technique of finding the relationship between two or more variables
- Change in dependent variable is associated with a change in one or more independent variables.

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2	
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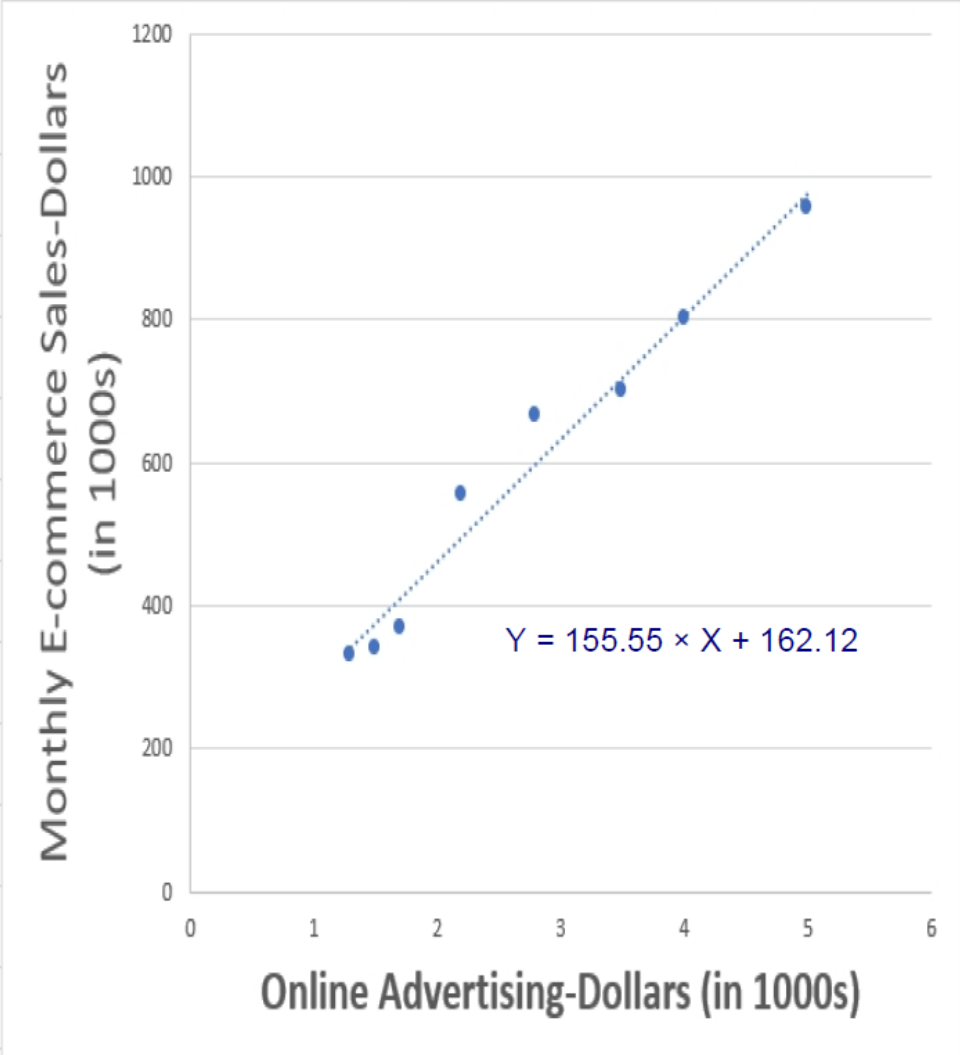
1.7	368
1.5	340
2.8	665
5	954
1.3	331
2.2	556
4	800
3.5	700
2	Y = 155.55 × 2 + 162.12 = 473.22
2.5	Y = 155.55 × 2.5 + 162.12 = 550.995
3	Y = 155.55 × 3 + 162.12 = 628.77



Regression Use Case

House Size (sq ft) x	Price ( ) y
800	1,500,000
1000	2,000,000
1200	2,500,000
1400	3,000,000
1600	3,500,000
1800	4,000,000
2000	4,500,000
2200	5,000,000
2400	5,500,000
2600	6,000,000
2800	6,500,000
3000	7,000,000

- Temperature vs. Number of cones sold at ice cream store
- Inches of rain vs. new cars sold
- Daily Snowfall vs. number of skier visits

If you think there is a relationship between two things regression would help confirm it!

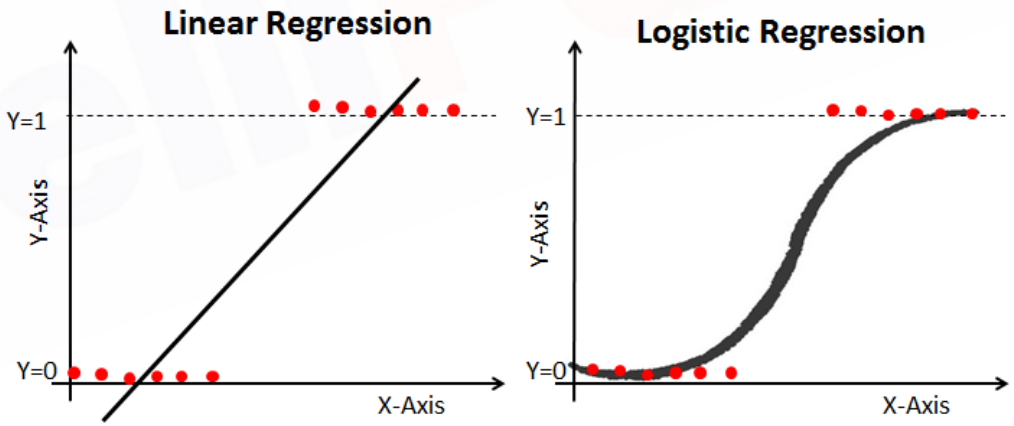
<<<<< $y = 2500x - 500000$

Another example:
Area(sq ft), #Bedrooms --> Price ( )

Types of Regression

There are various types of Regression, but we will focus on:

- Linear Regression
- Logistic Regression

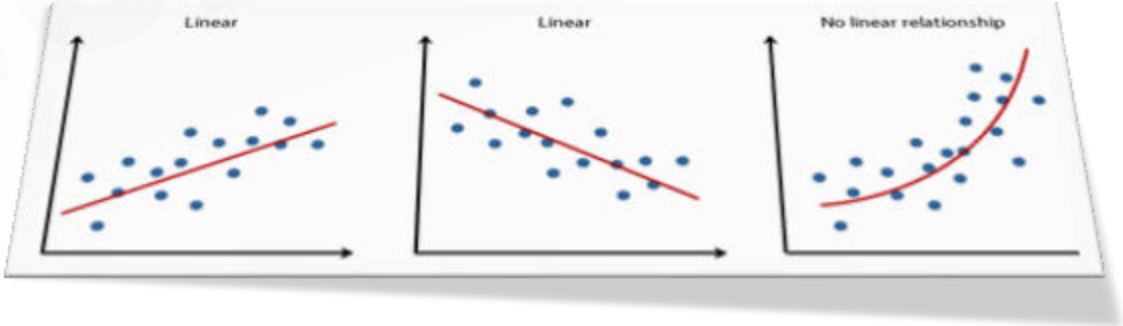


Types of Regression

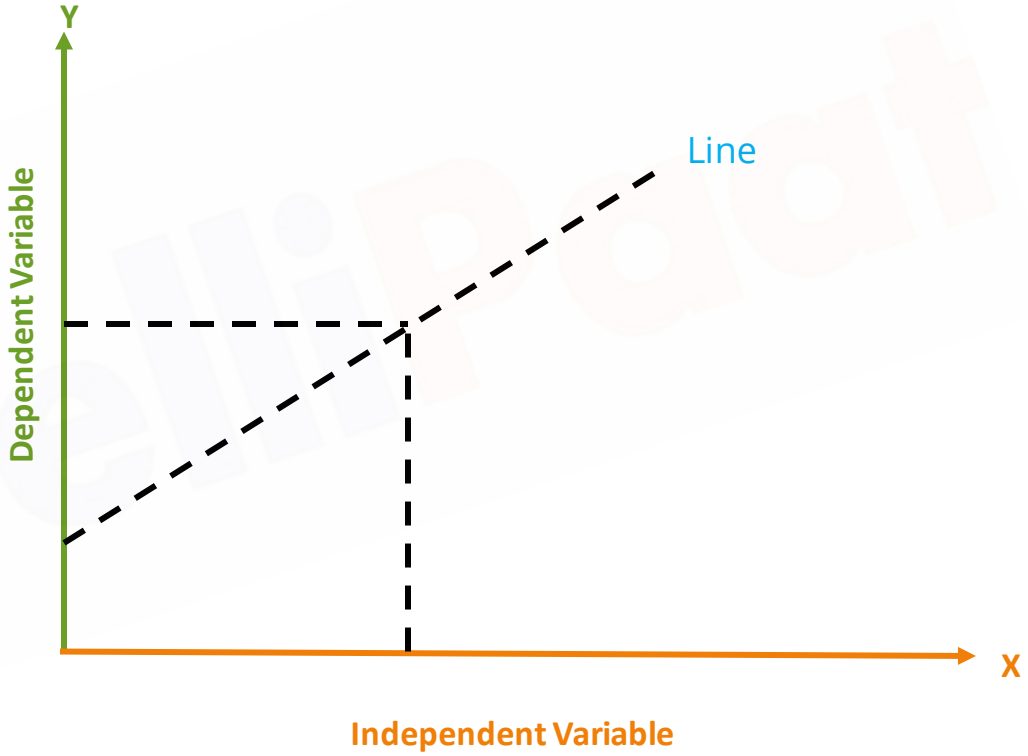
LINEAR REGRESSION	LOGISTIC REGRESSION
Continuous Variables	Categorical Variables
Solves Regression Issue	Solves Classification Issue
Straight Line	S-Curve

What is Linear Regression?

- Simple linear regression is useful for finding relationship between two continuous variables
- One is predictor or independent variable and other is response or dependent variable



Understanding Linear Regression



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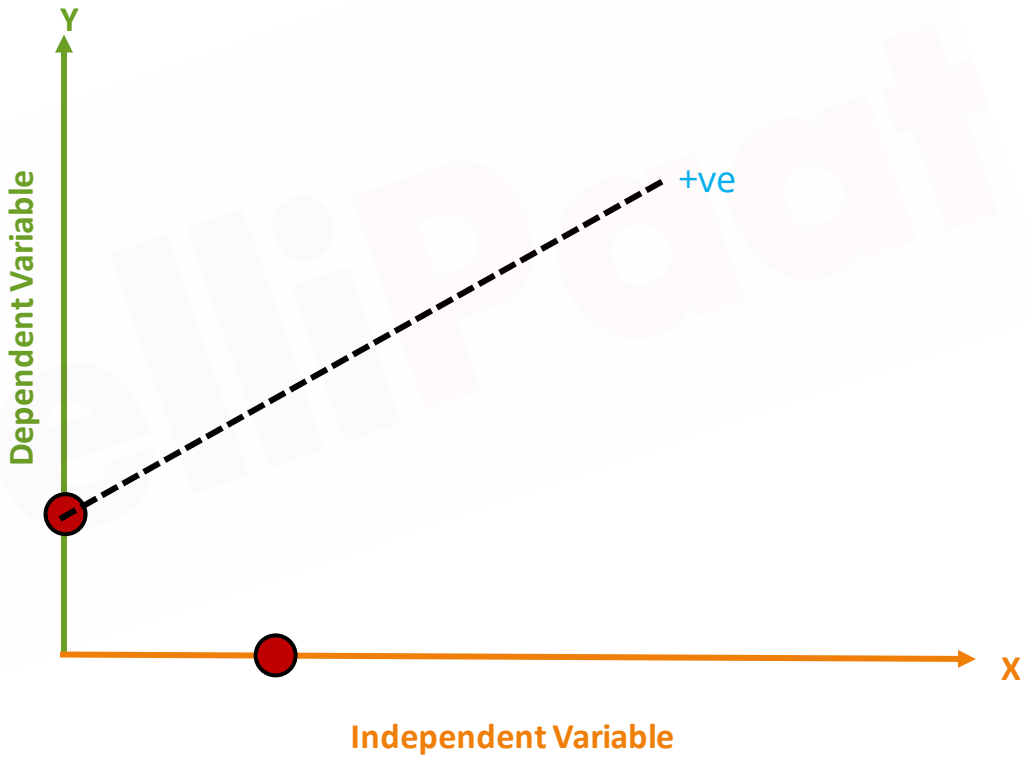

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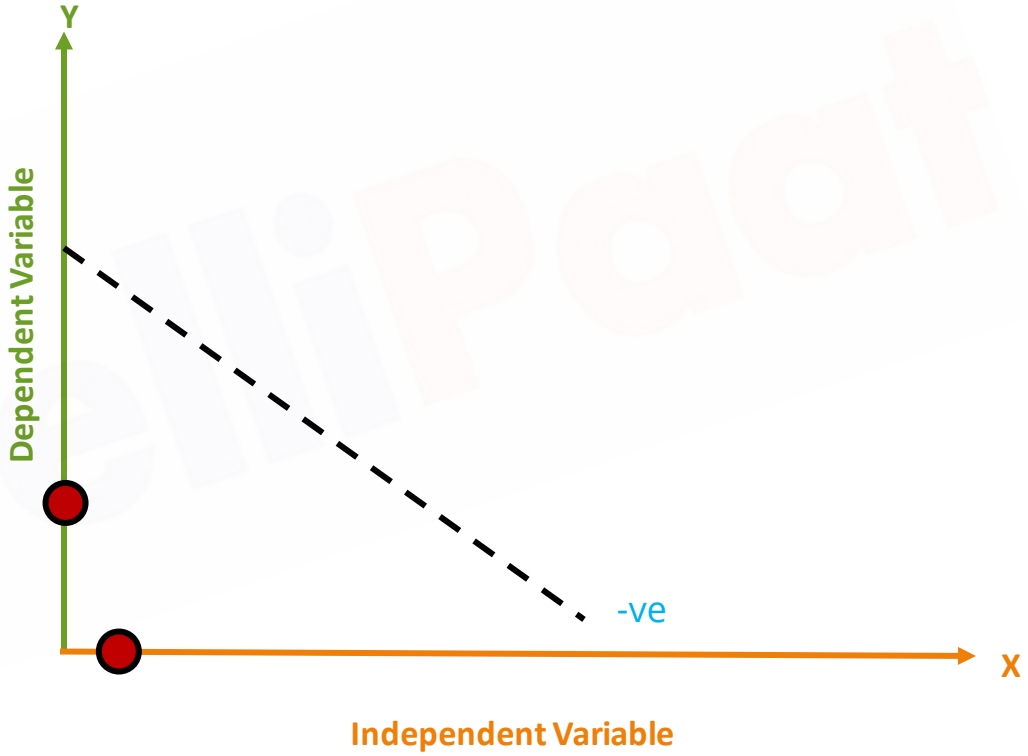

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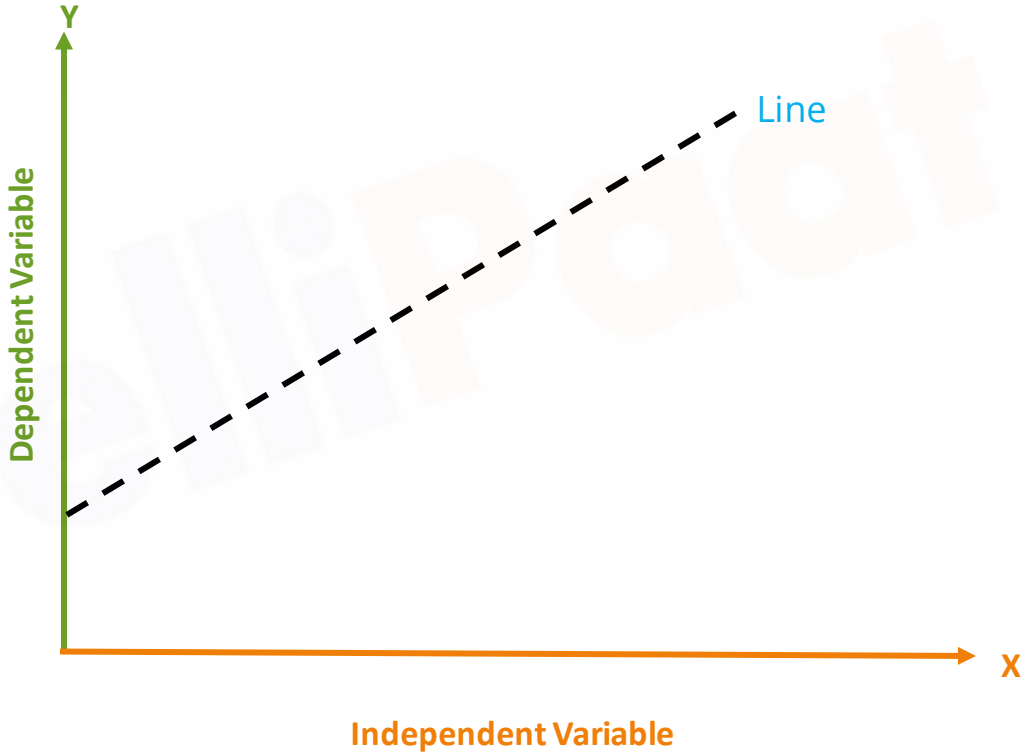

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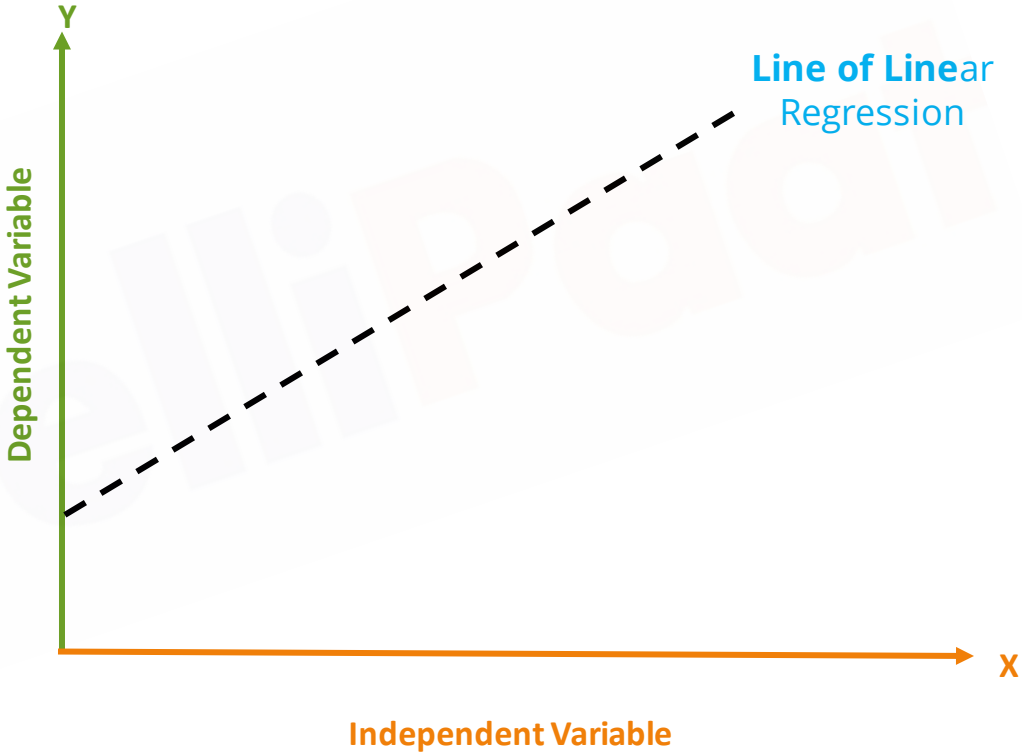

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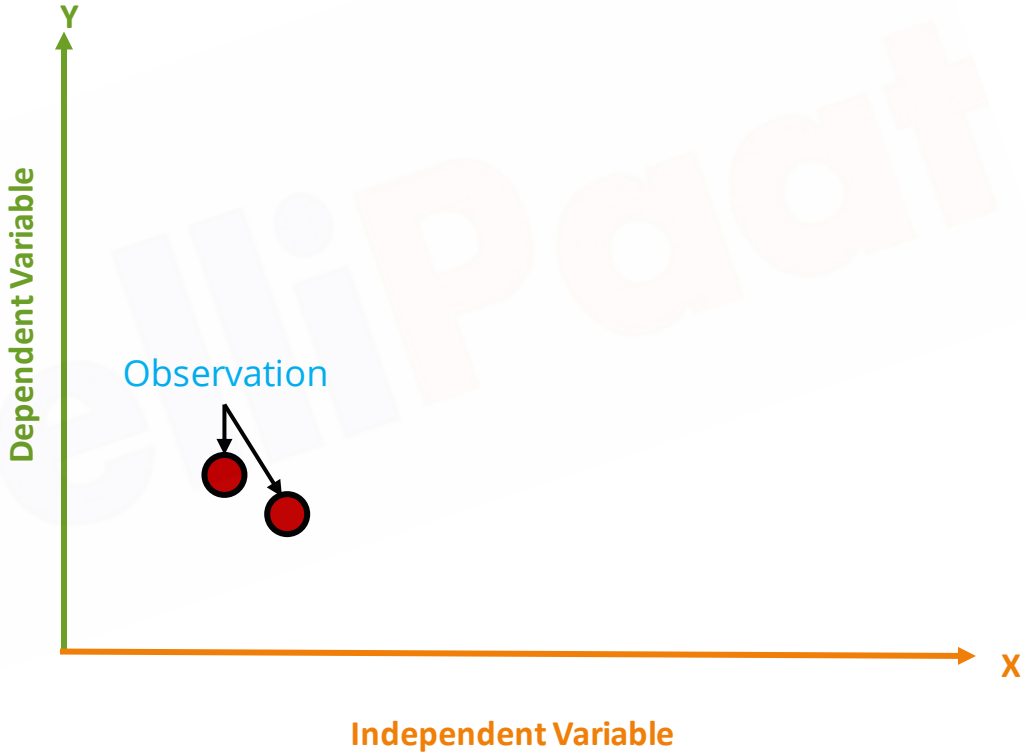

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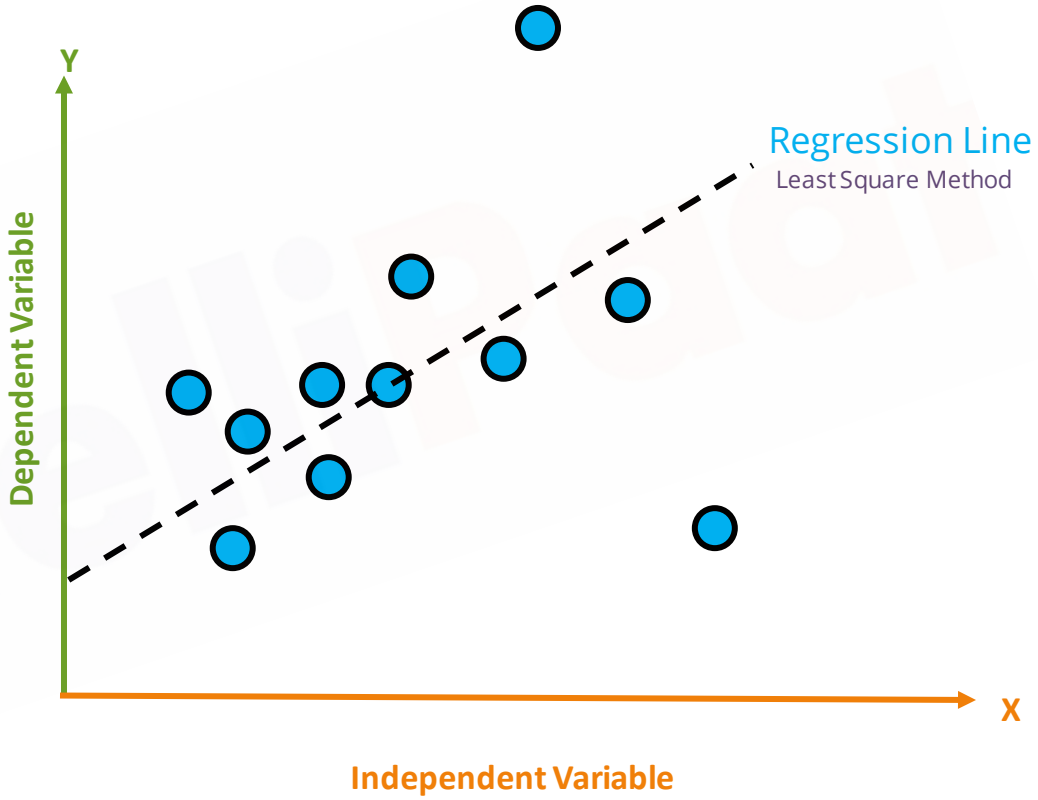

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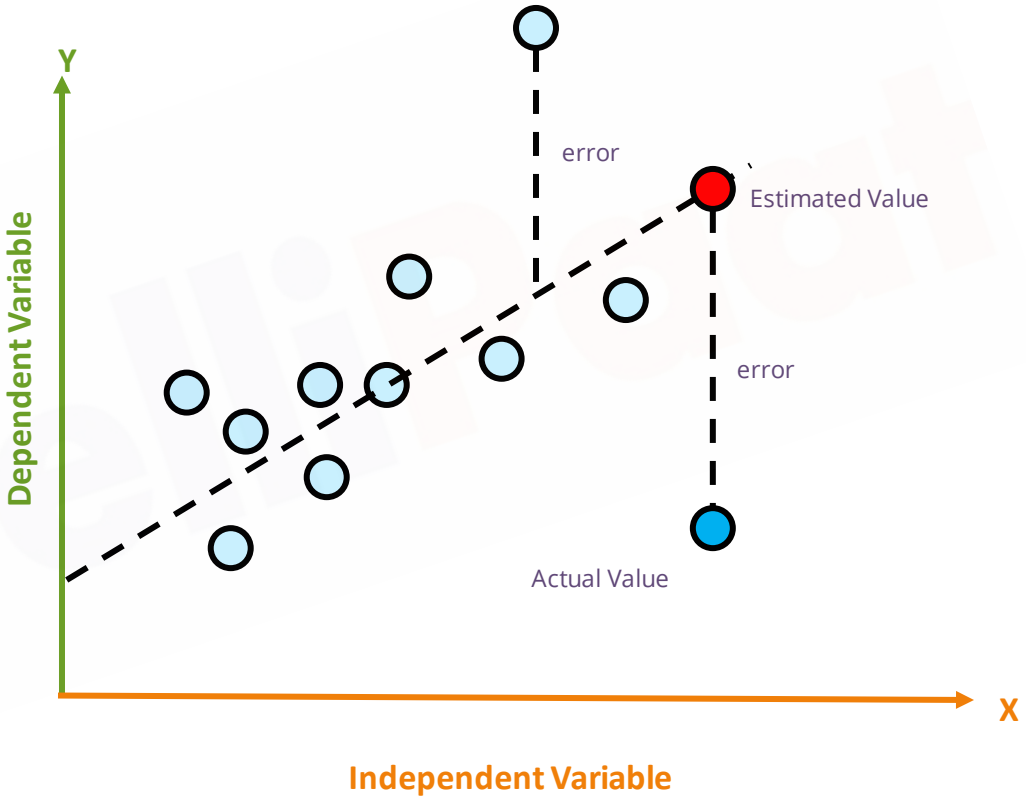

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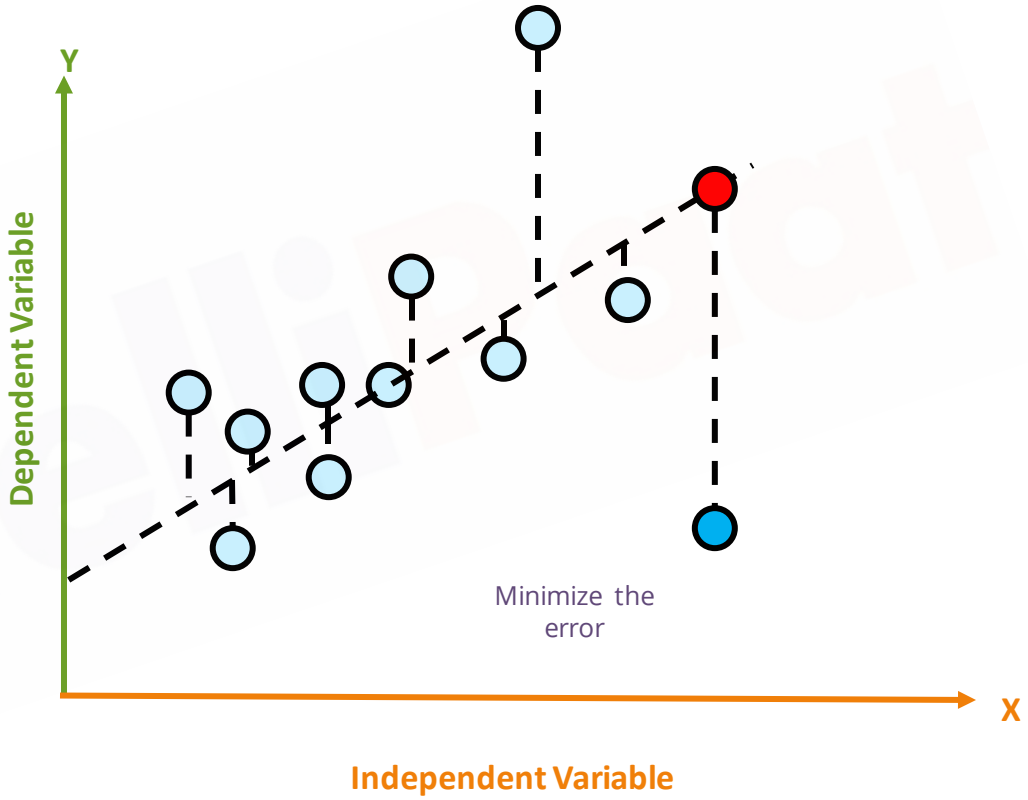

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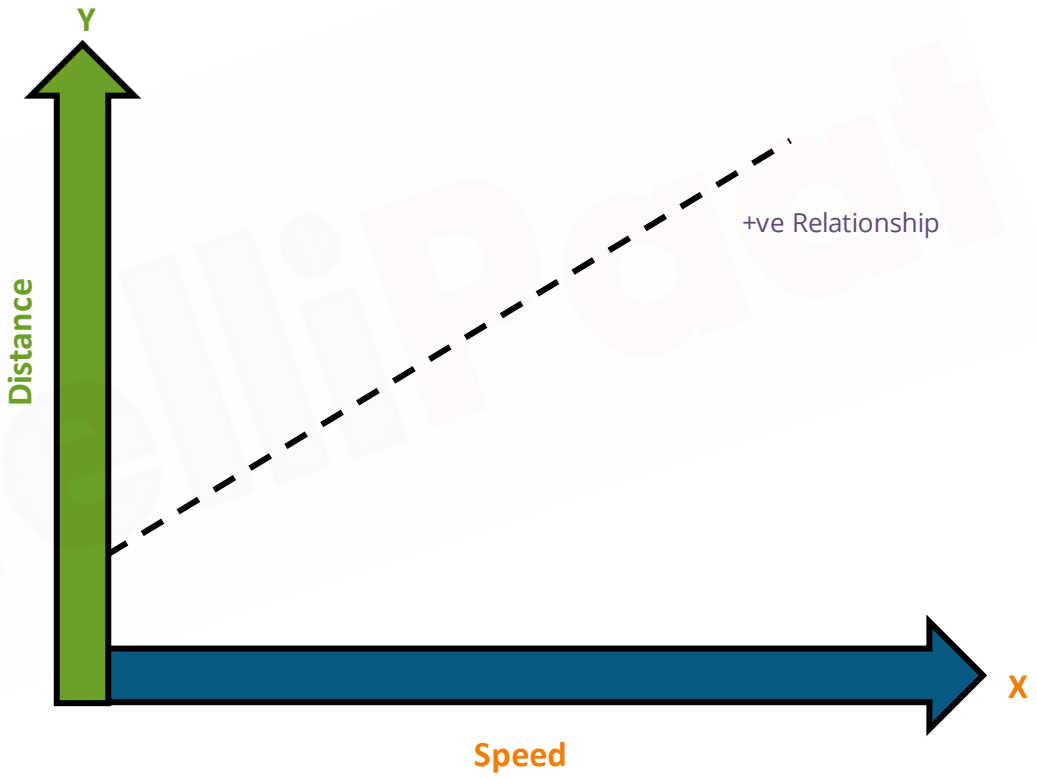

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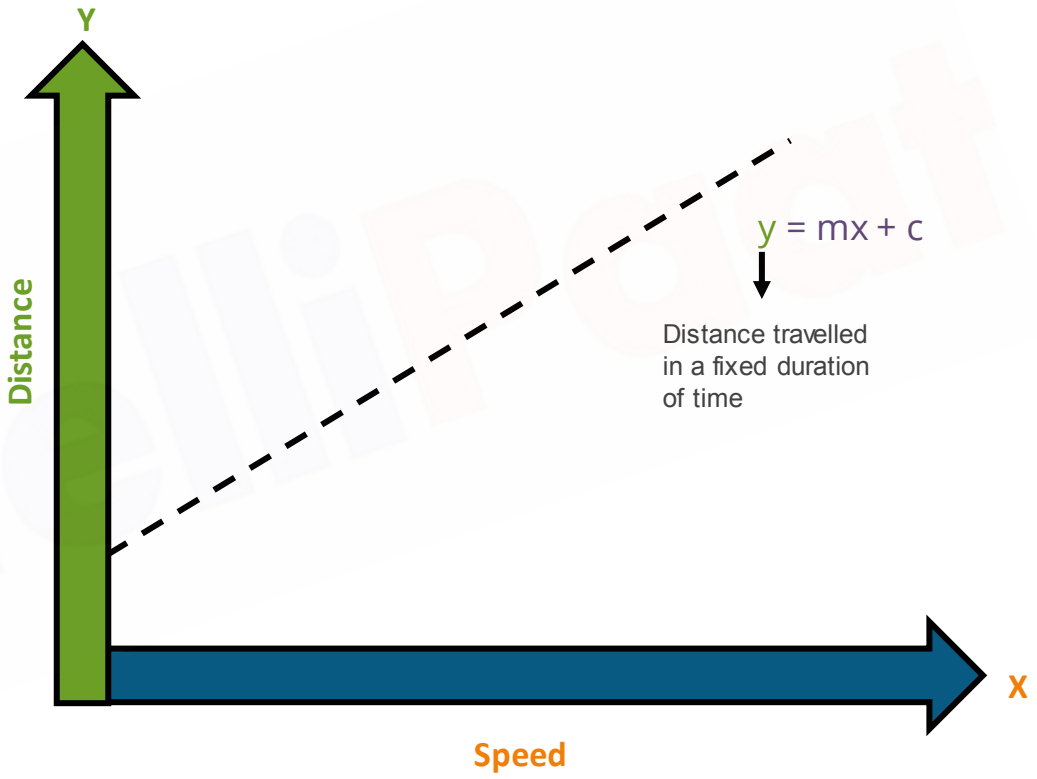

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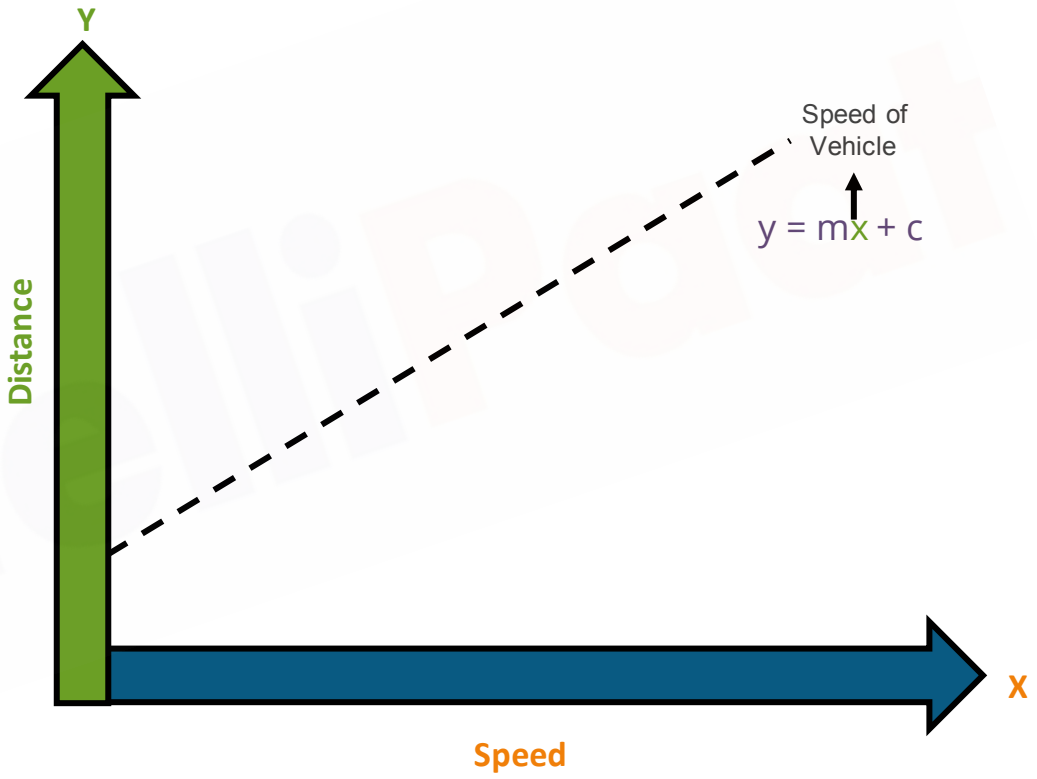

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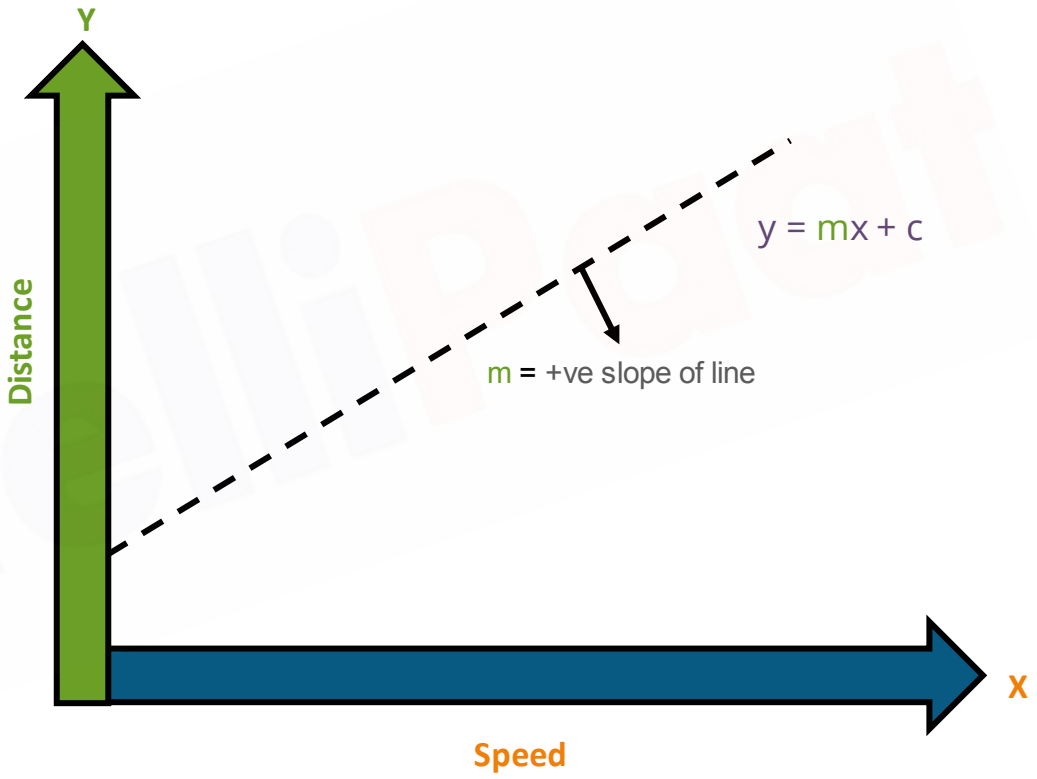

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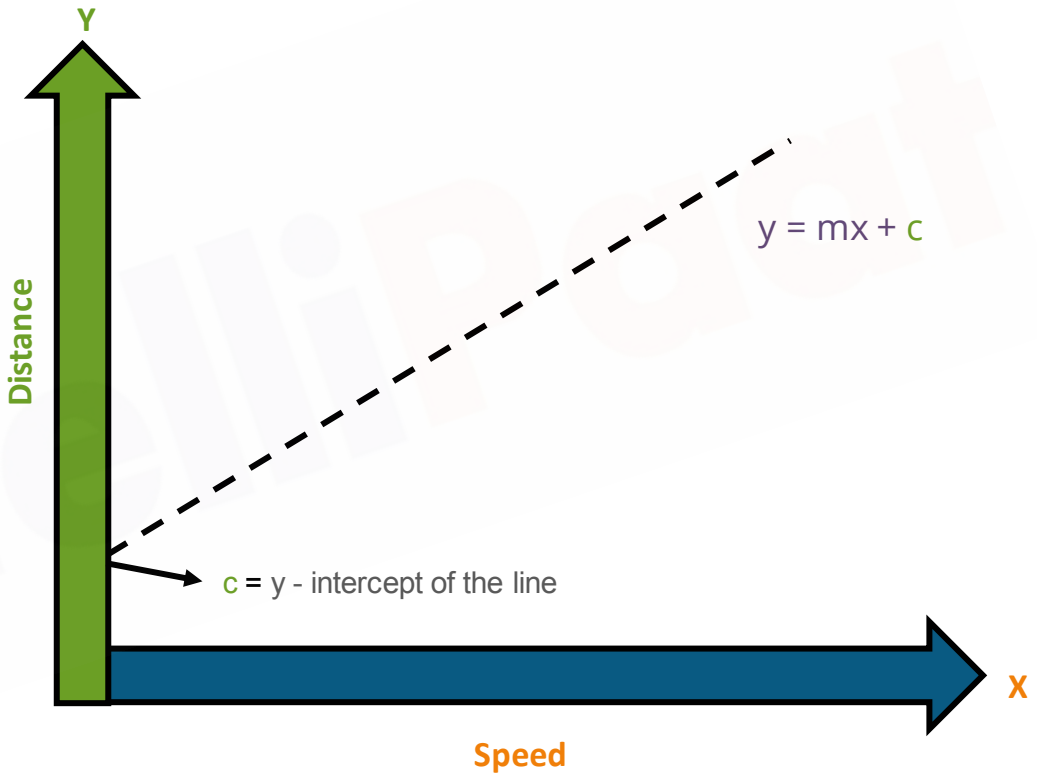

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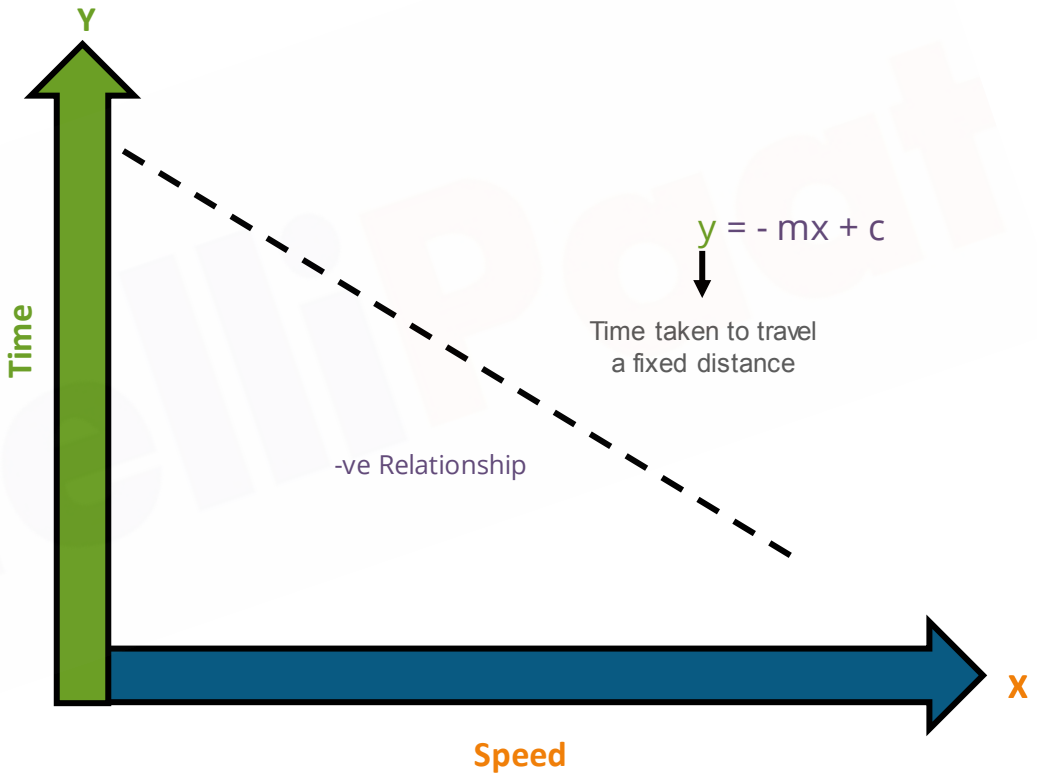

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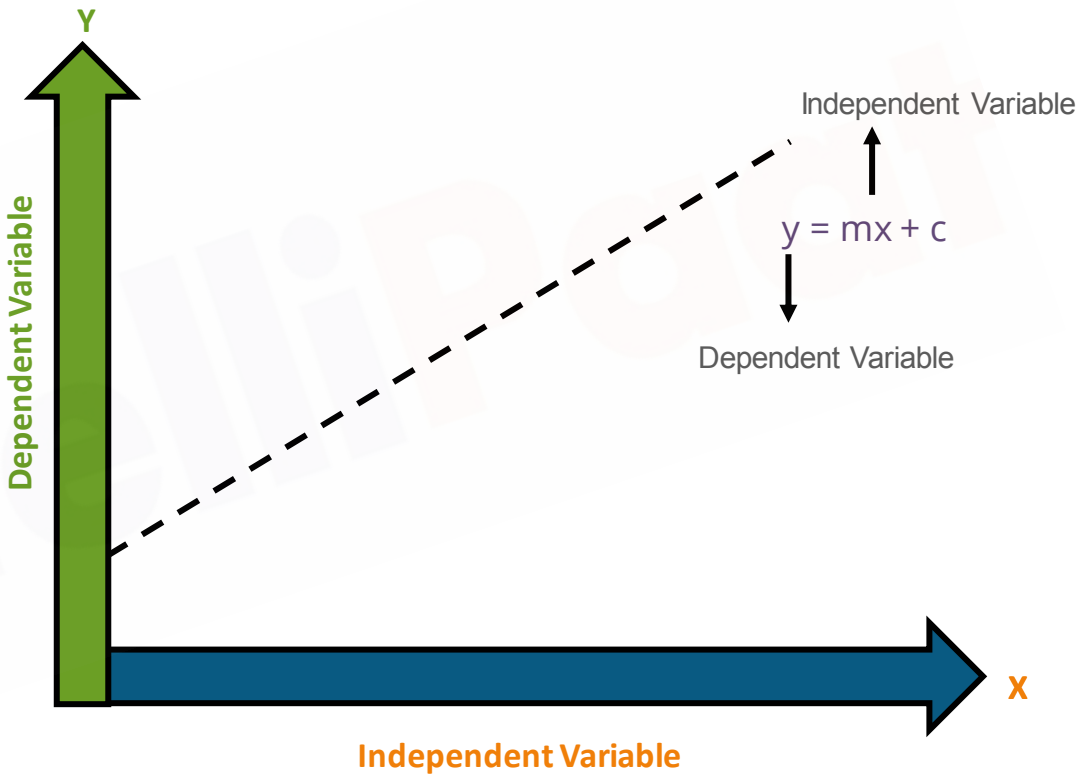
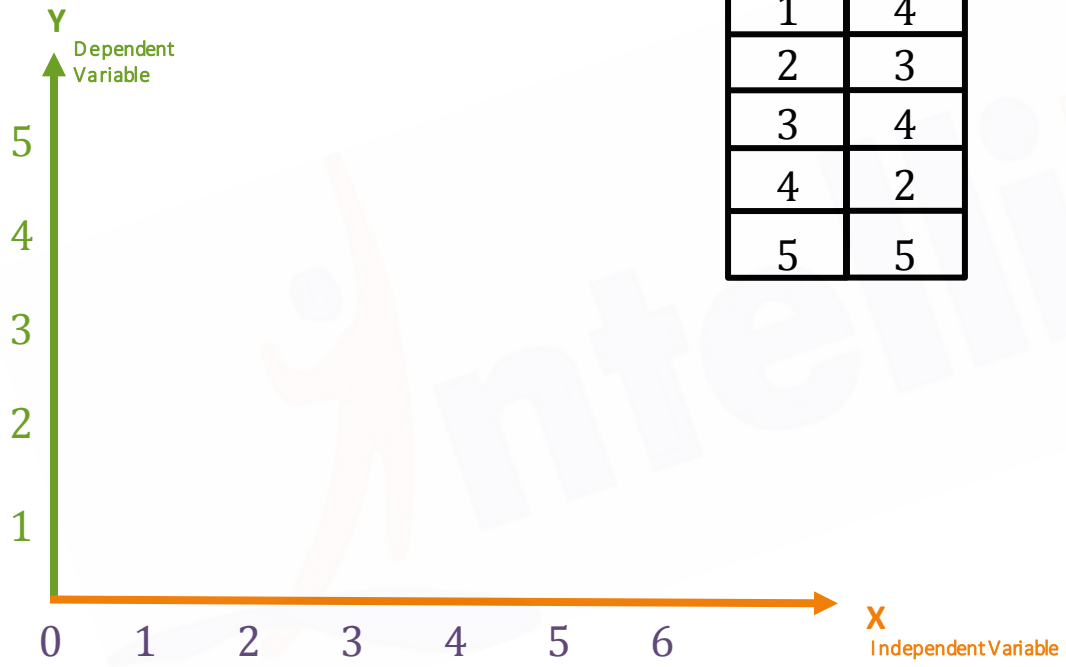


Understanding Linear Regression

x	y
1	4
2	3
3	4
4	2
5	5



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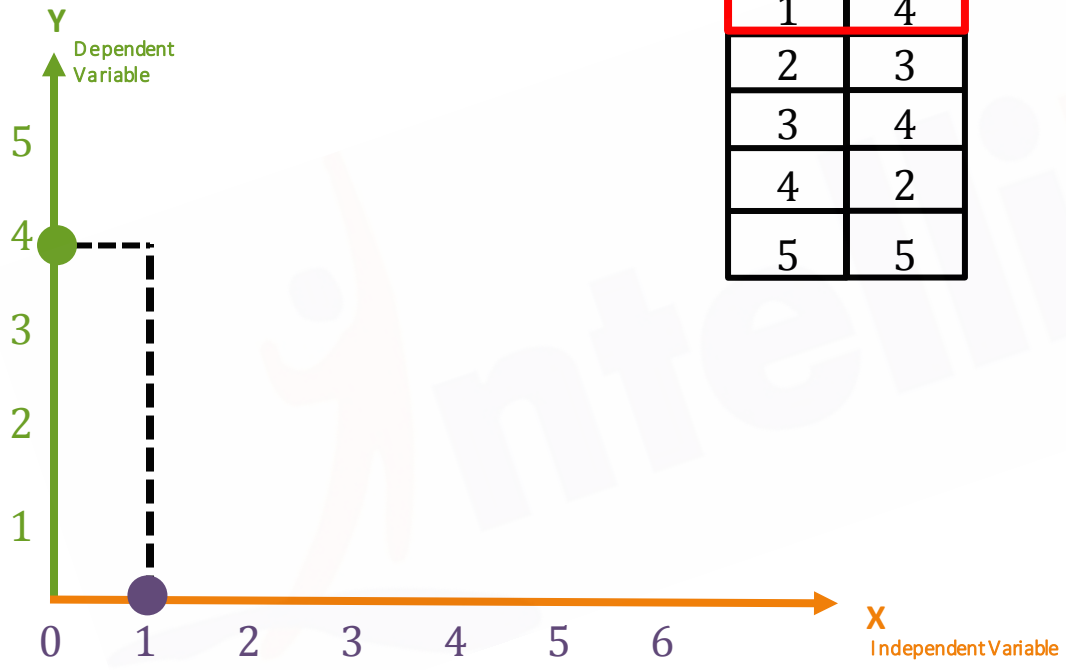

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x	y
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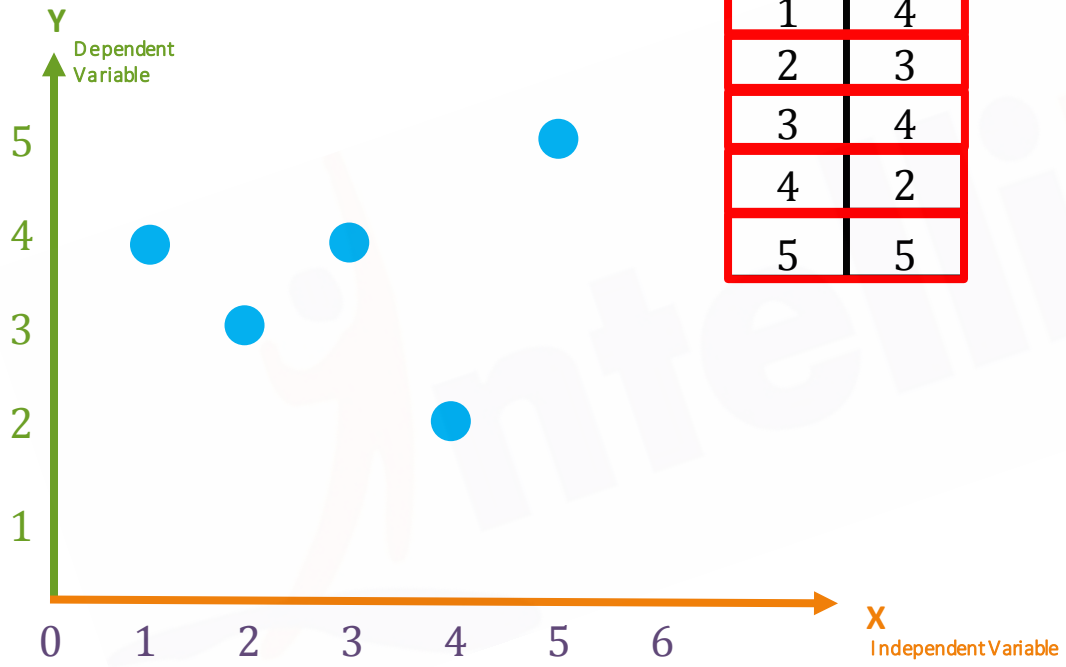
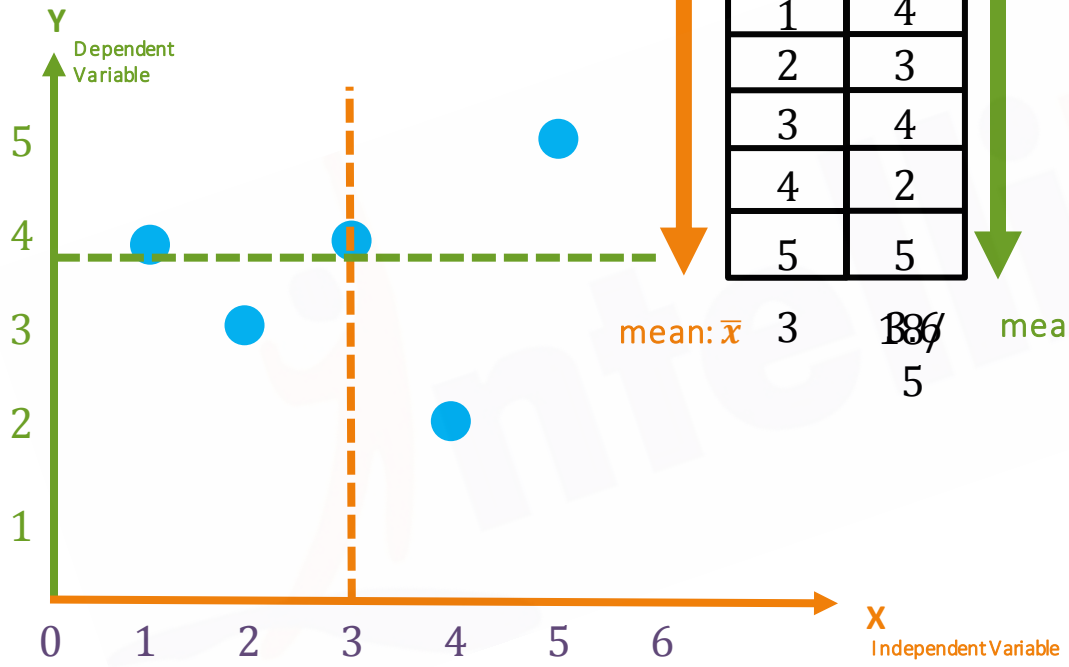


Understanding Linear Regression

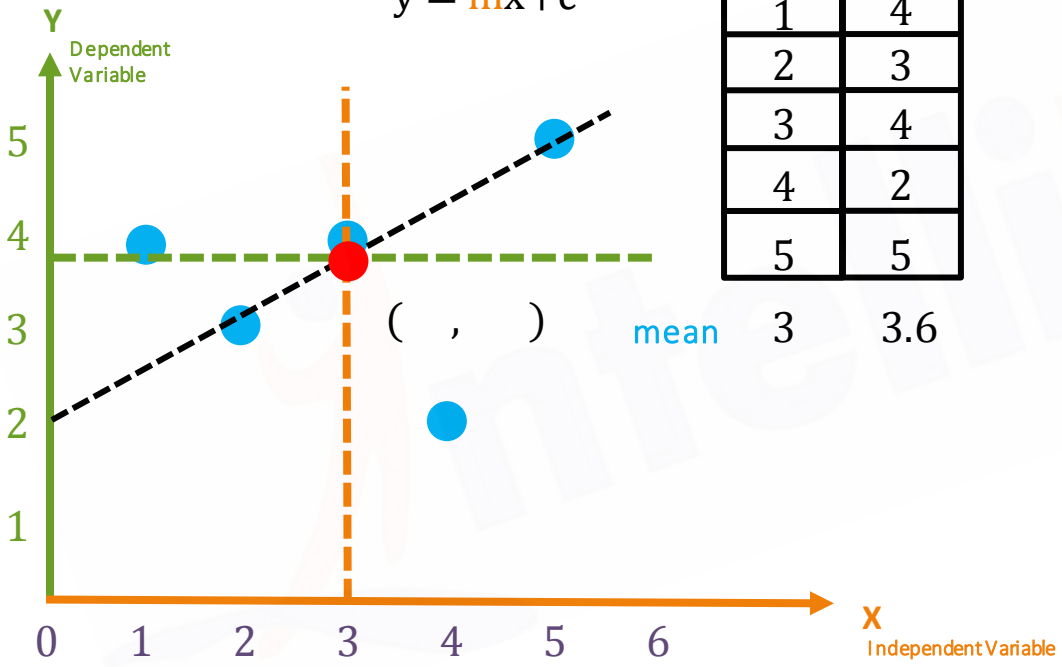
x	y
1	4
2	3
3	4
4	2
5	5

mean: $\bar{x}$ 3 mean : $\bar{y}$ $\frac{18}{5}$ 3.6



Understanding Linear Regression

$$y = mx + c$$



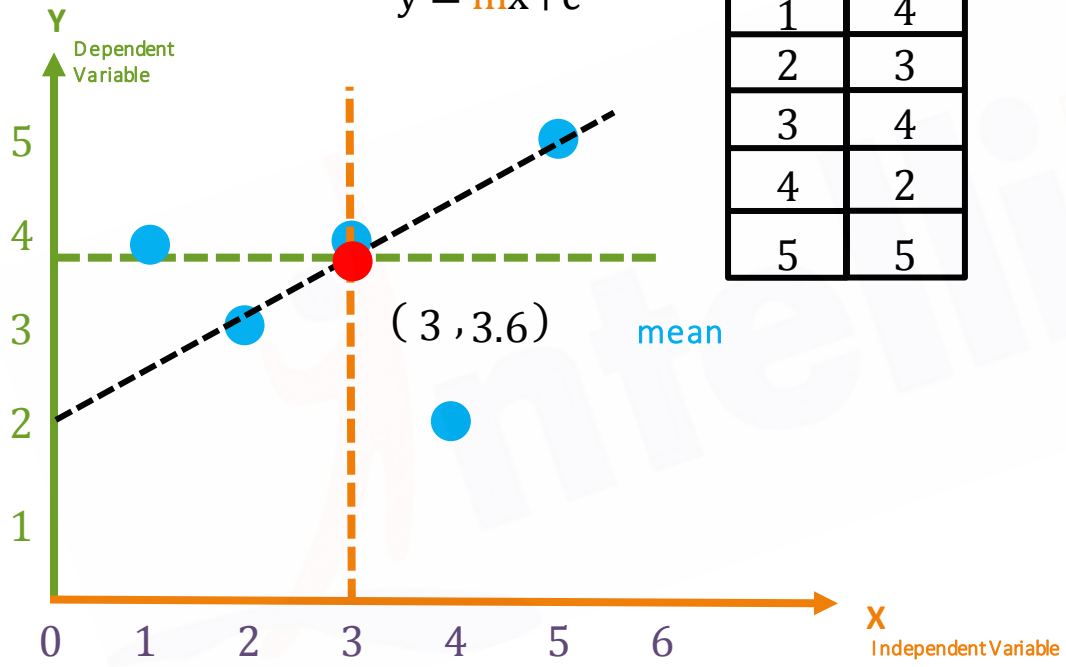
x	y
1	4
2	3
3	4
4	2
5	5
mean 3	3.6

(,)

Understanding Linear Regression

$$y = mx + c$$

x	y
1	4
2	3
3	4
4	2
5	5



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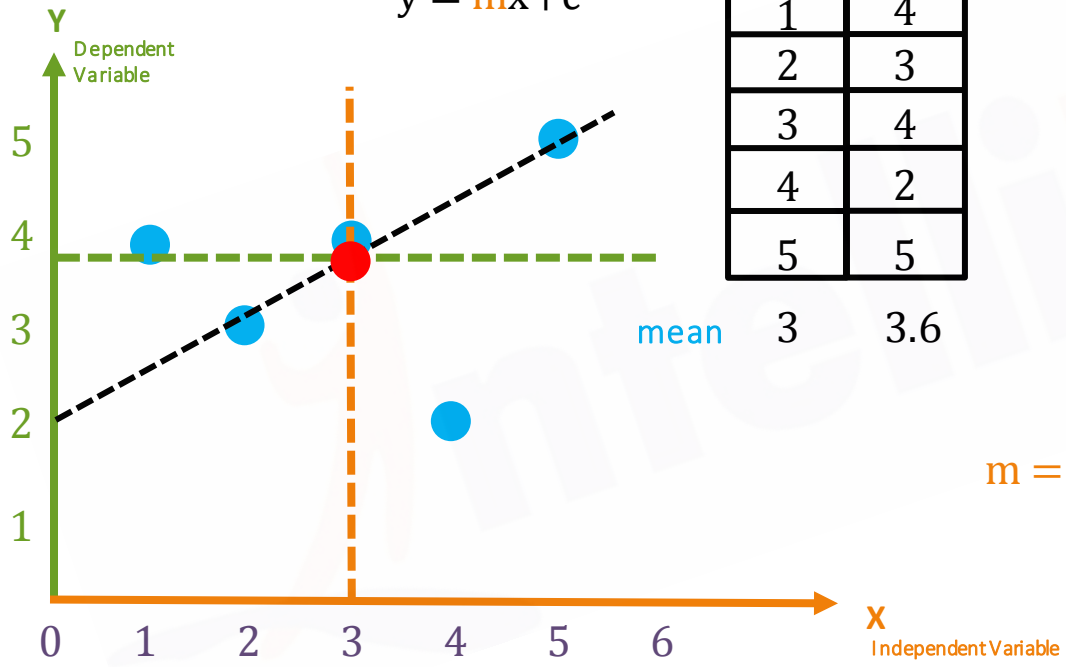
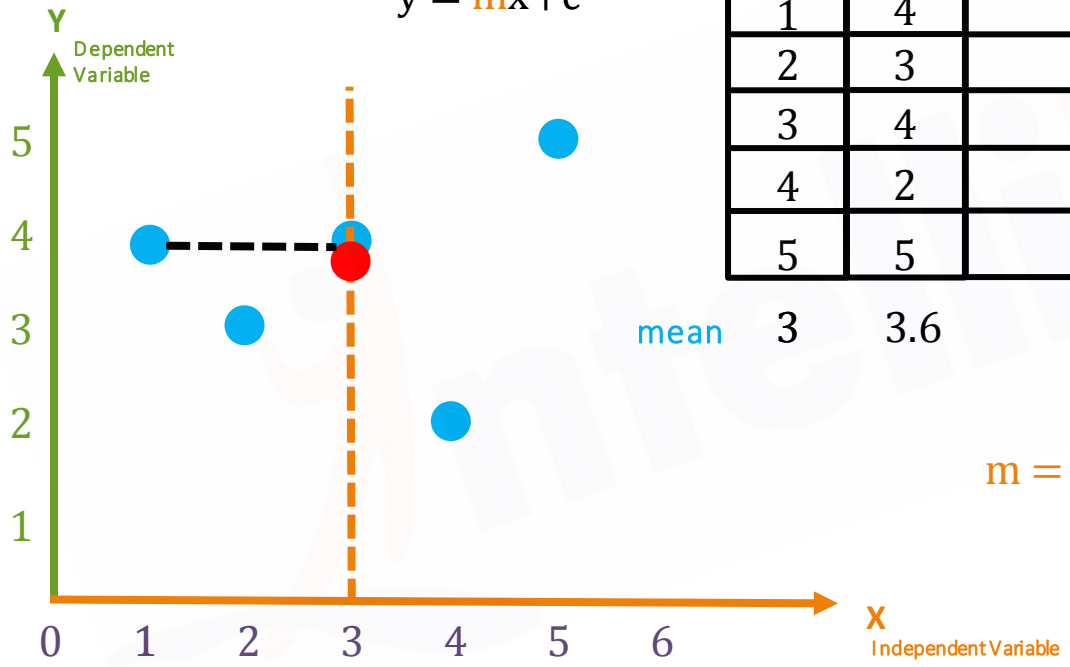


x	y
1	4
2	3
3	4
4	2
5	5
mean 3	3.6

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$



x	y	$x - \bar{x}$
1	4	
2	3	
3	4	
4	2	
5	5	
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

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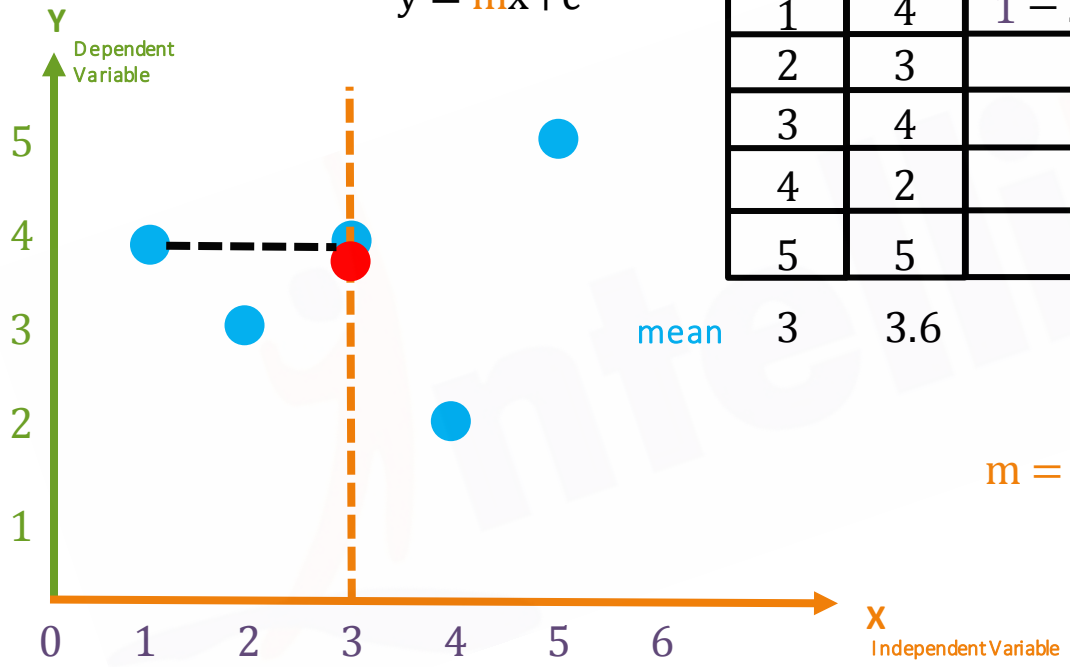

x	y	$x - \bar{x}$
1	4	$1 - 3$
2	3	
3	4	
4	2	
5	5	
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

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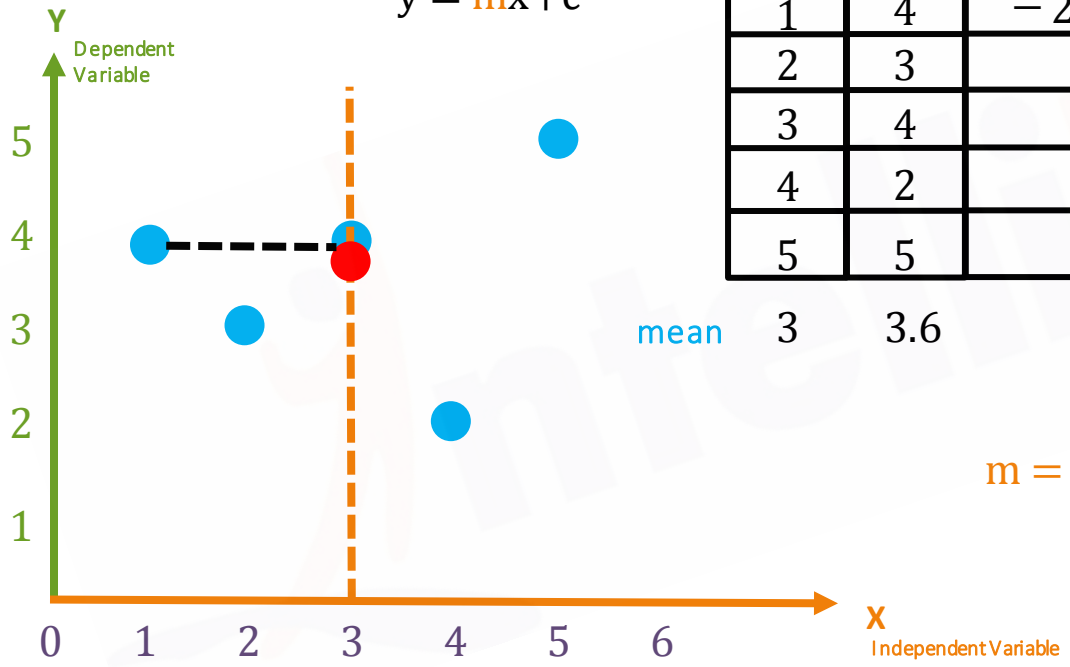
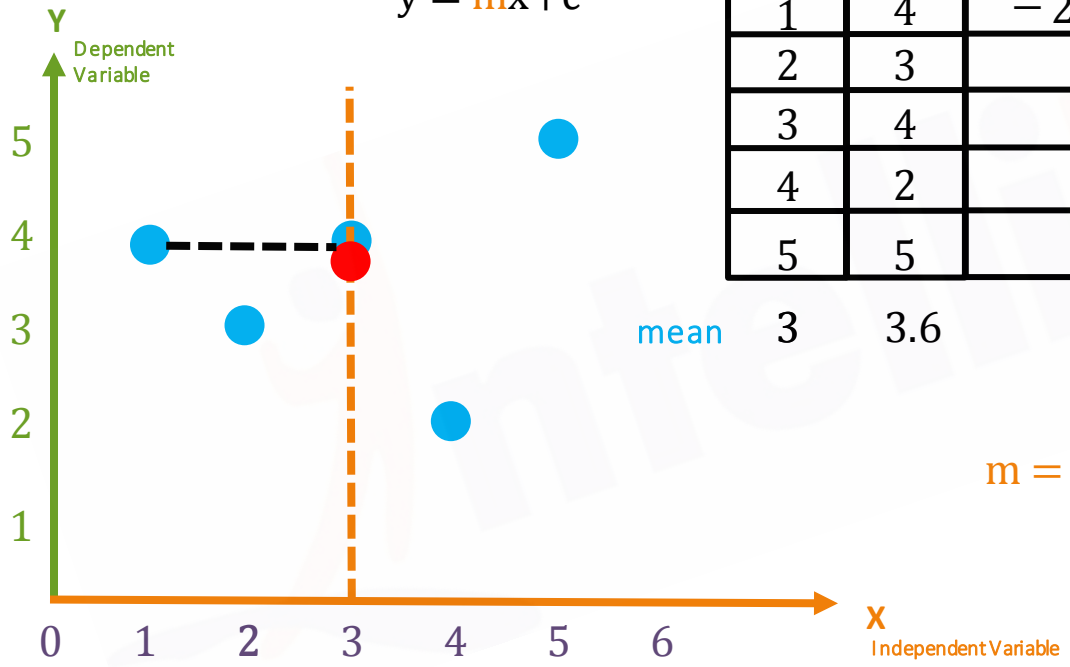


x	y	$x - \bar{x}$
1	4	-2
2	3	
3	4	
4	2	
5	5	
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$



x	y	$x - \bar{x}$
1	4	-2
2	3	
3	4	
4	2	
5	5	
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

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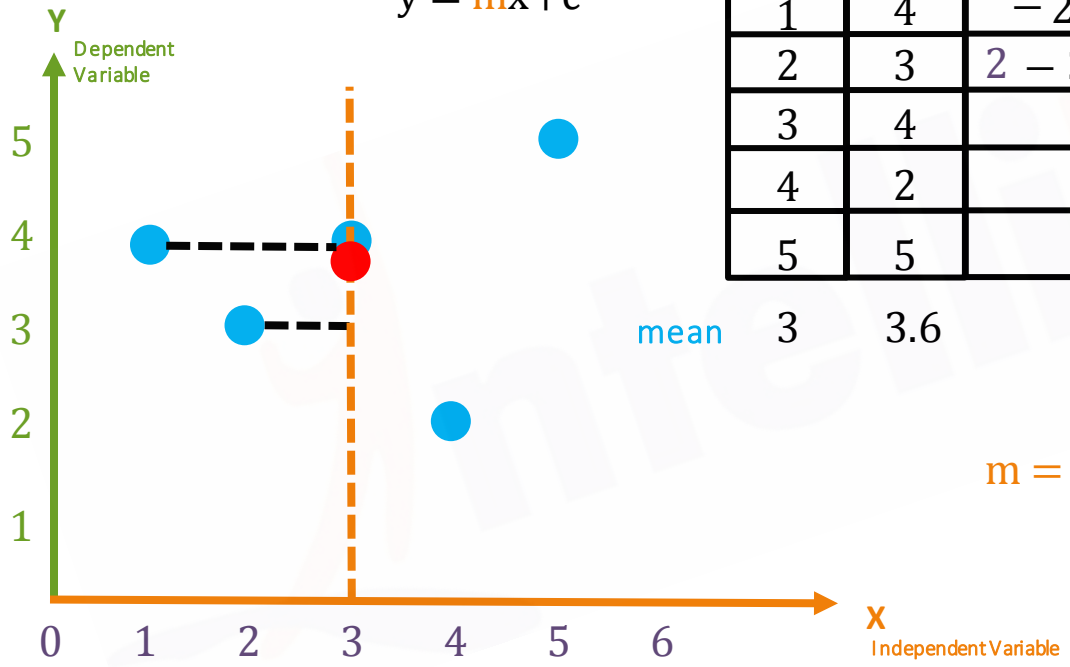
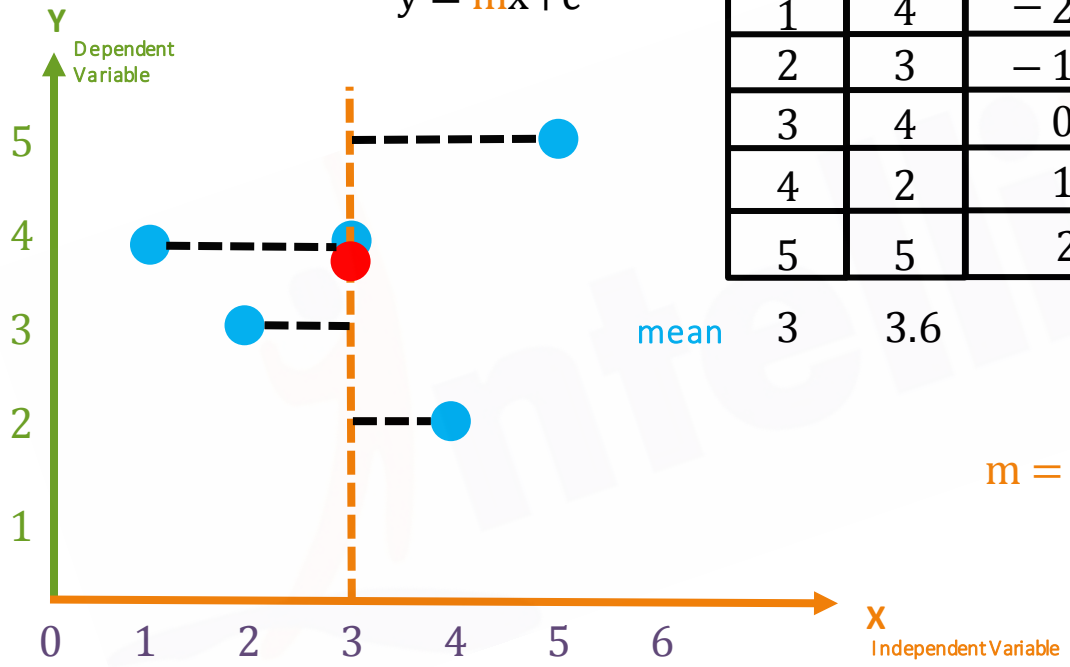


x	y	$x - \bar{x}$
1	4	-2
2	3	$2 - 3$
3	4	
4	2	
5	5	
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

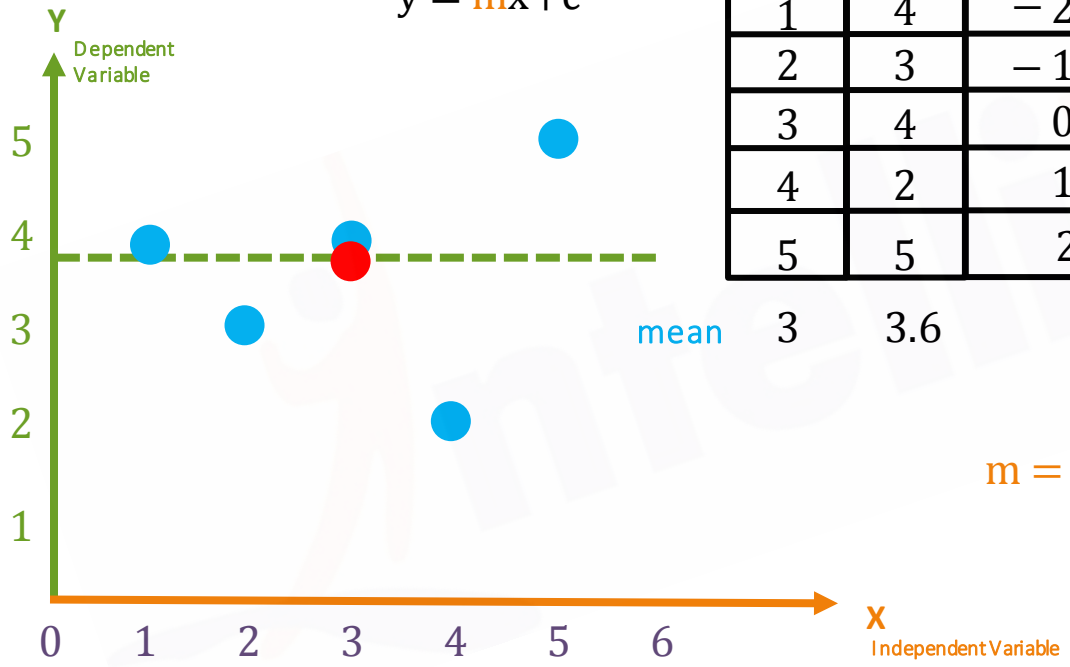


x	y	$x - \bar{x}$
1	4	-2
2	3	-1
3	4	0
4	2	1
5	5	2
mean 3	3.6	

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

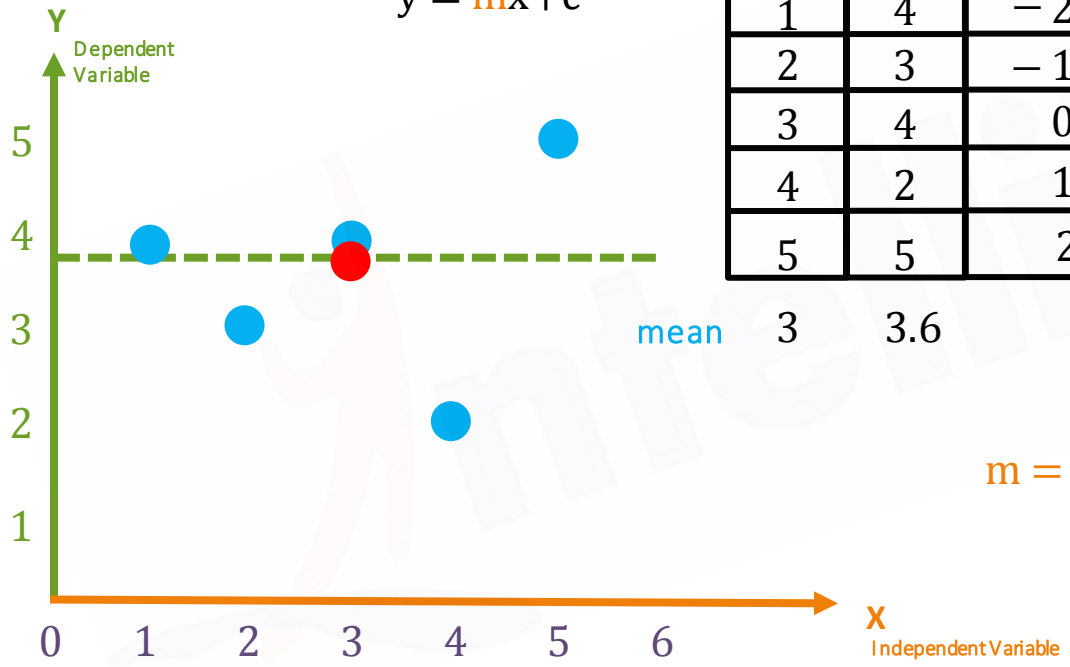


x	y	$x-\bar{x}$	$y-\bar{y}$
1	4	-2	
2	3	-1	
3	4	0	
4	2	1	
5	5	2	
mean 3	3.6		

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

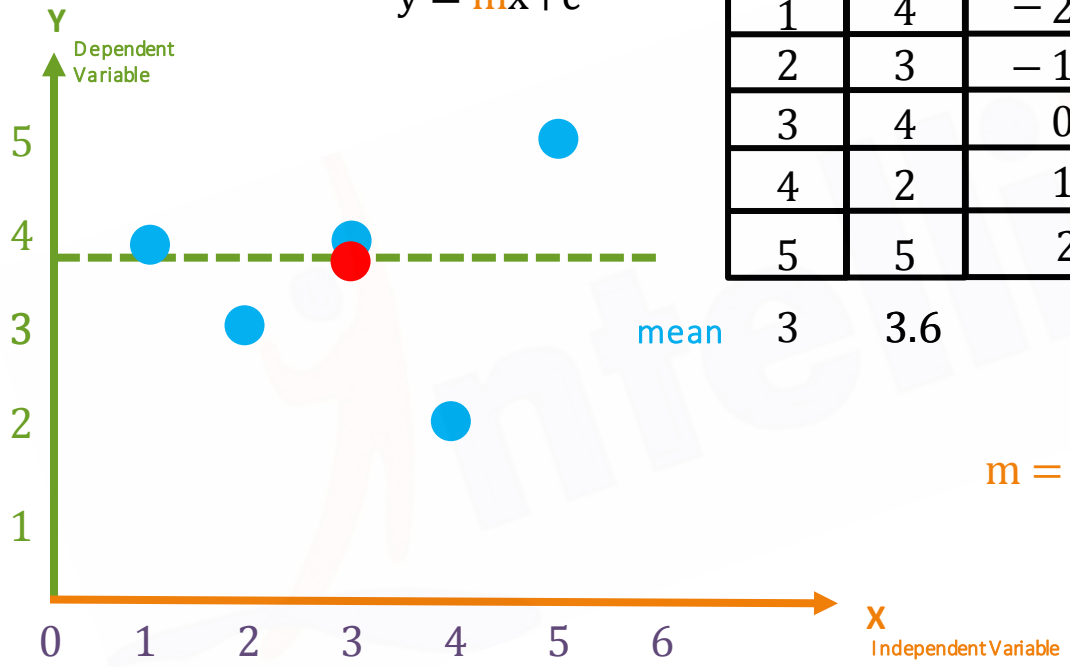


x	y	$x-\bar{x}$	$y-\bar{y}$
1	4	-2	$4-3.6$
2	3	-1	
3	4	0	
4	2	1	
5	5	2	
mean	3	3.6	

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

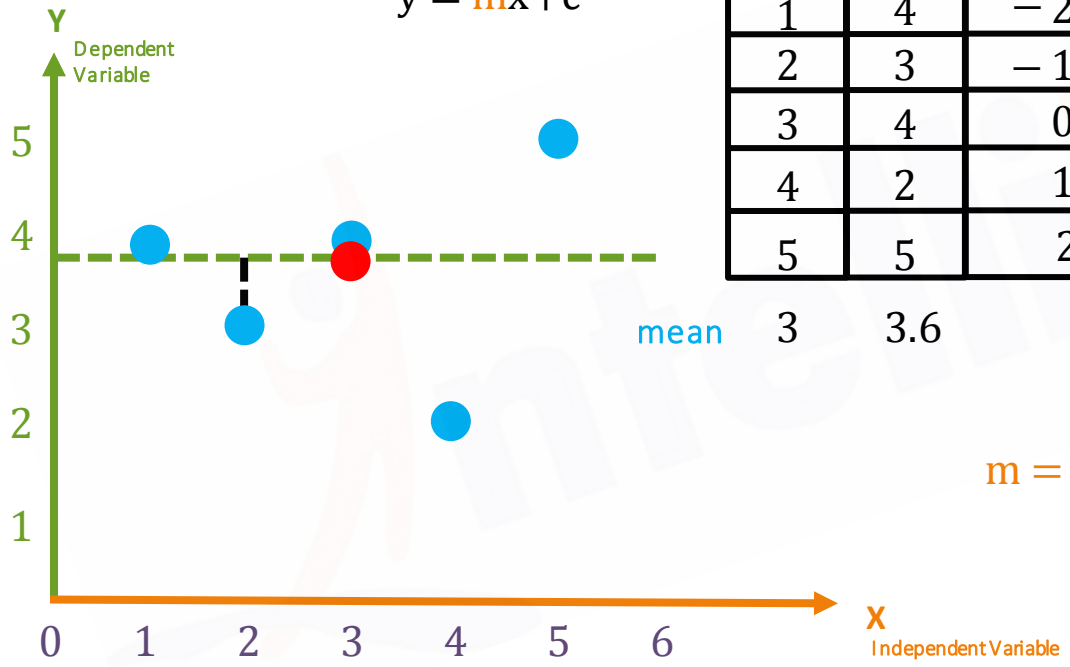


x	y	$x-\bar{x}$	$y-\bar{y}$
1	4	-2	0.4
2	3	-1	
3	4	0	
4	2	1	
5	5	2	
mean 3	3.6		

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$

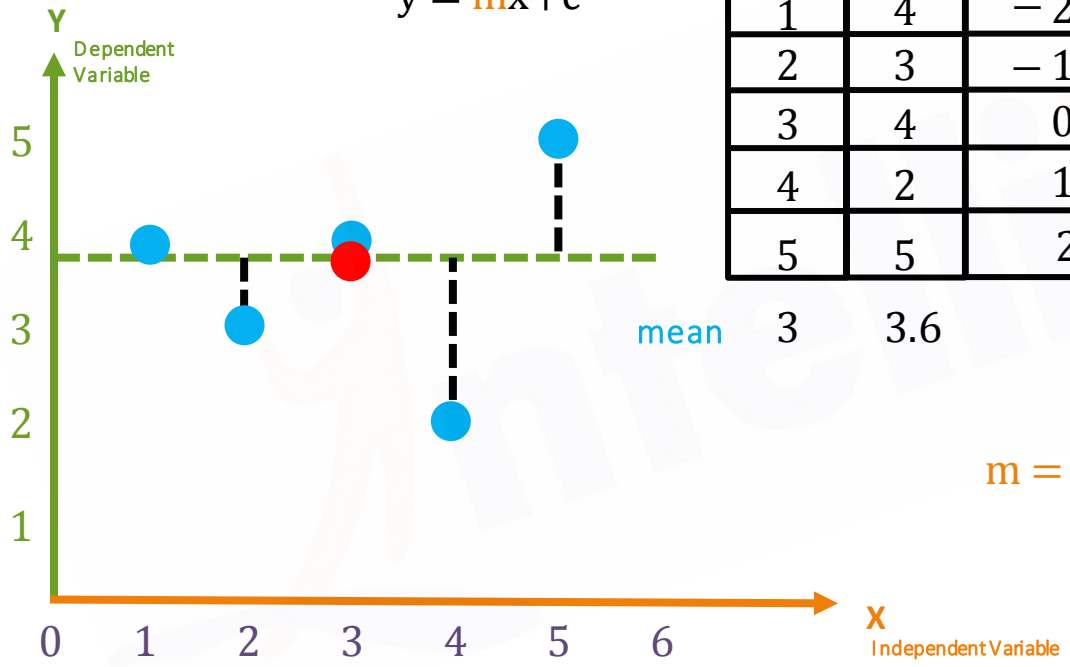


x	y	$x - \bar{x}$	$y - \bar{y}$
1	4	-2	0.4
2	3	-1	$3 - 3.6$
3	4	0	
4	2	1	
5	5	2	
mean 3	3.6		

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

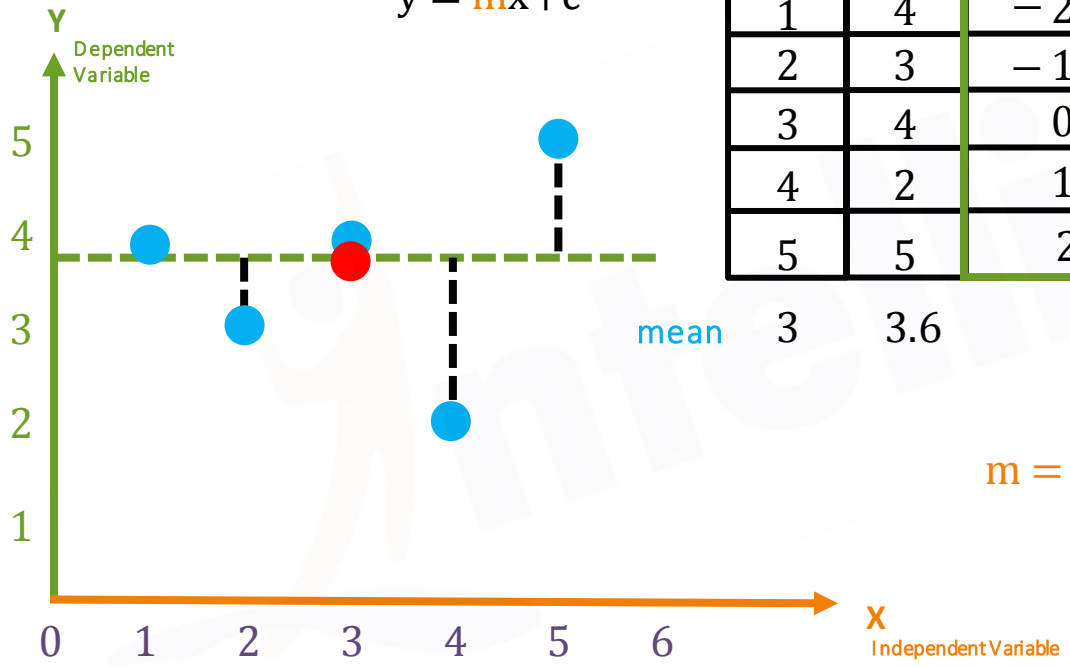


x	y	$x-\bar{x}$	$y-\bar{y}$
1	4	-2	0.4
2	3	-1	-0.6
3	4	0	0.4
4	2	1	-1.6
5	5	2	1.4
mean 3	3.6		

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

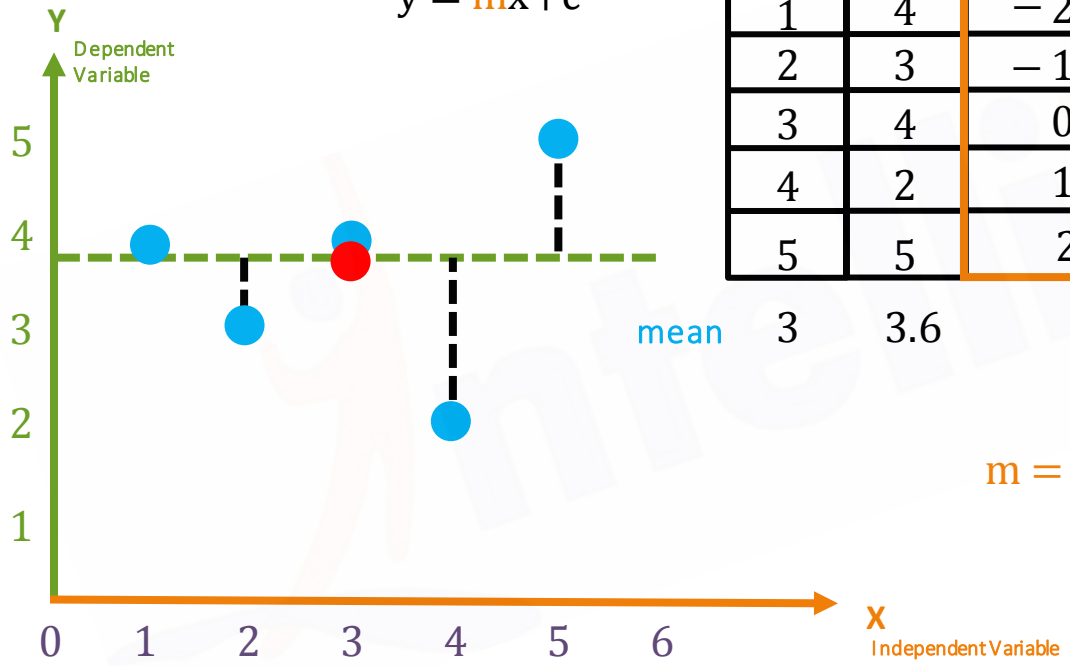


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$
1	4	-2	0.4	4
2	3	-1	-0.6	1
3	4	0	0.4	0
4	2	1	-1.6	1
5	5	2	1.4	4
mean	3		3.6	

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$

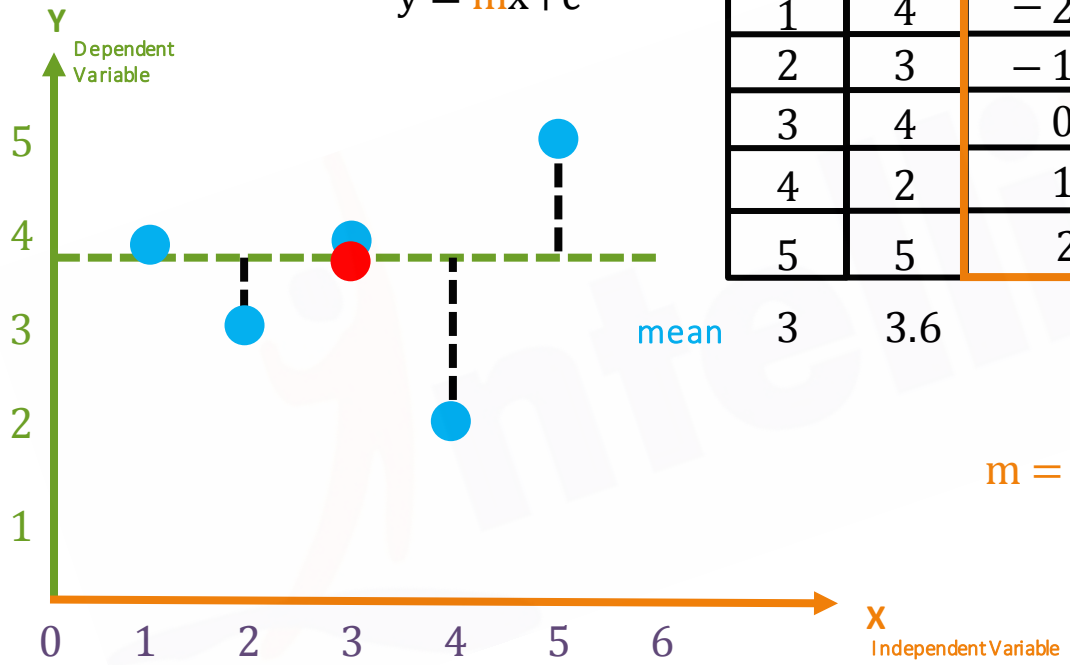


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-2 x 0.4
2	3	-1	-0.6	1	
3	4	0	0.4	0	
4	2	1	-1.6	1	
5	5	2	1.4	4	
mean	3	3.6			

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$

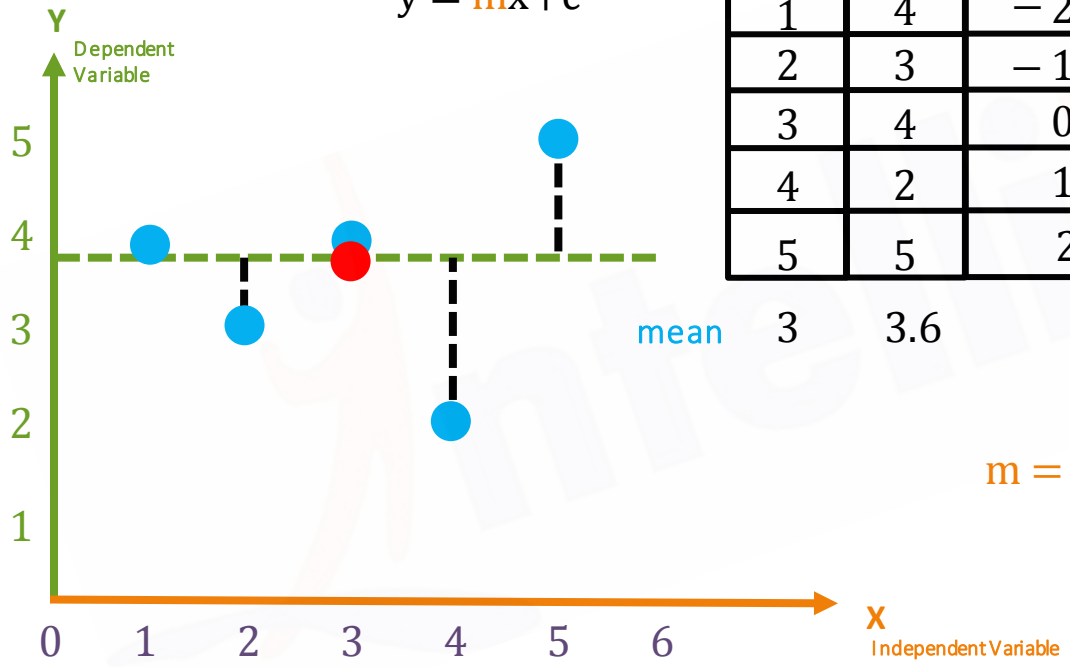


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6			

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Understanding Linear Regression

$y = mx + c$

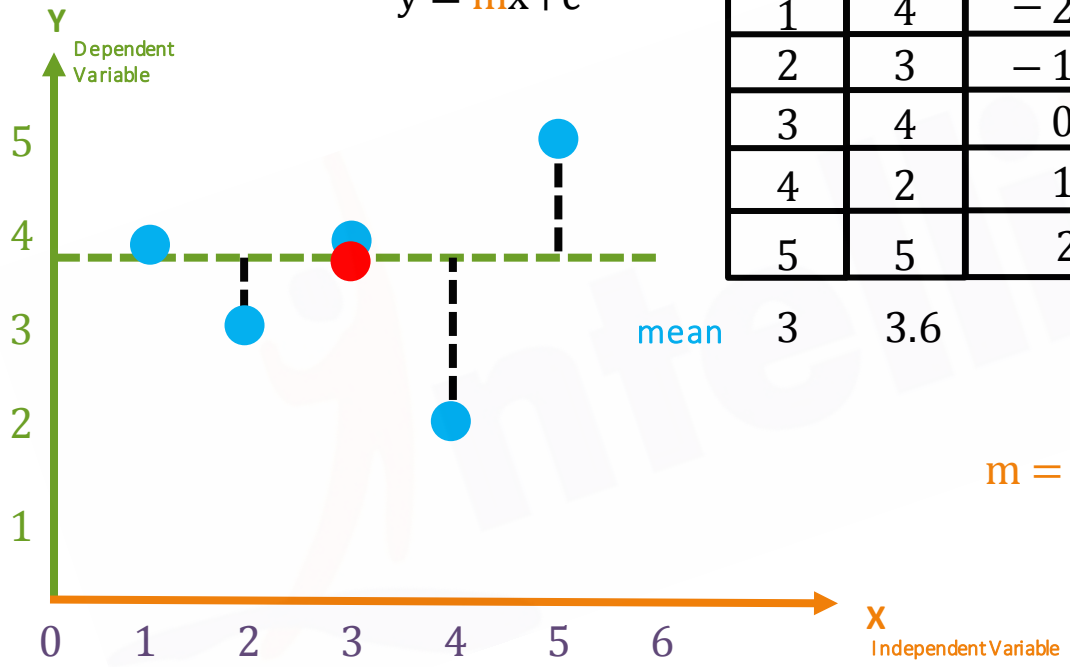


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2}$$

Understanding Linear Regression

$$y = mx + c$$

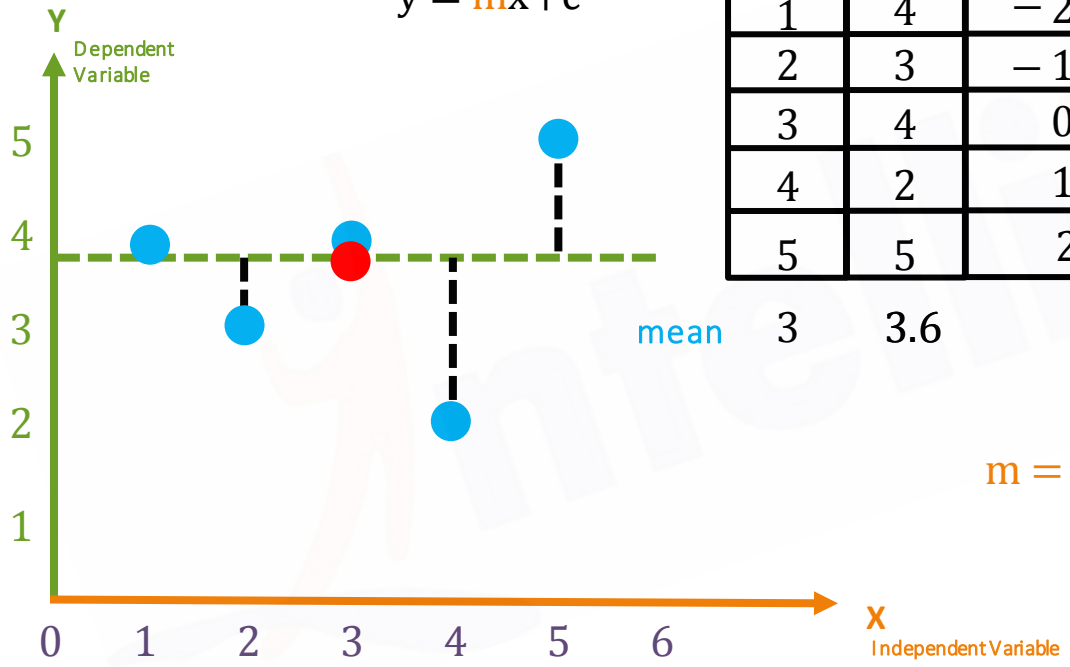


x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2} = \frac{1}{10}$$

Understanding Linear Regression

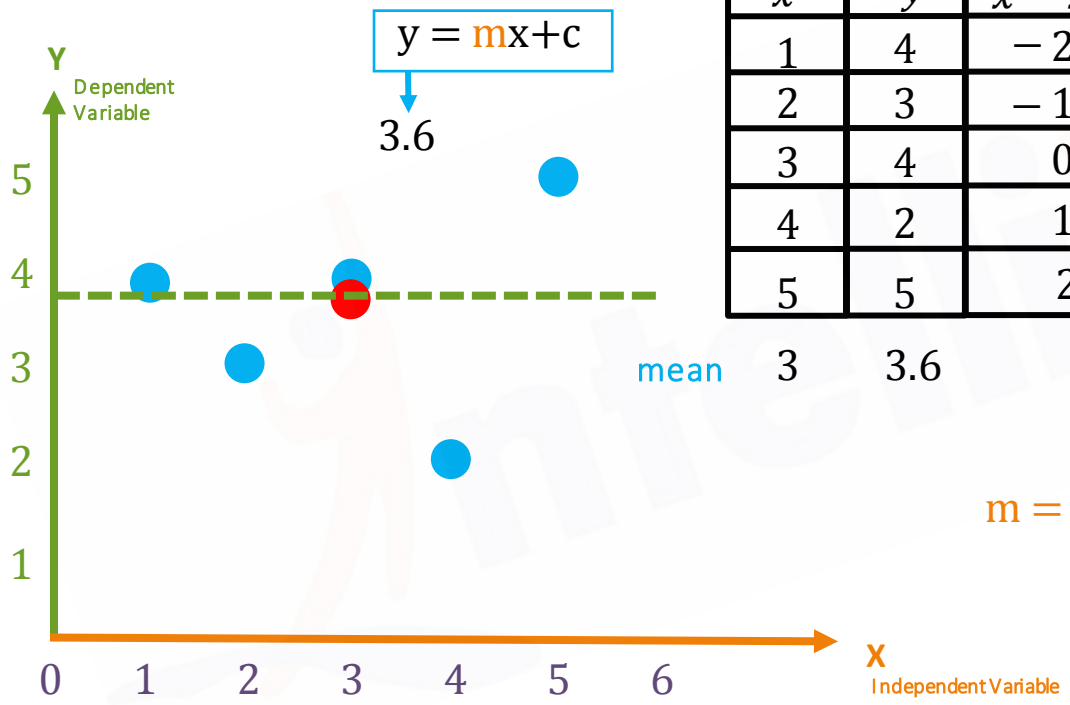
$y = mx + c$



x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2} = \mathbf{0.1}$$

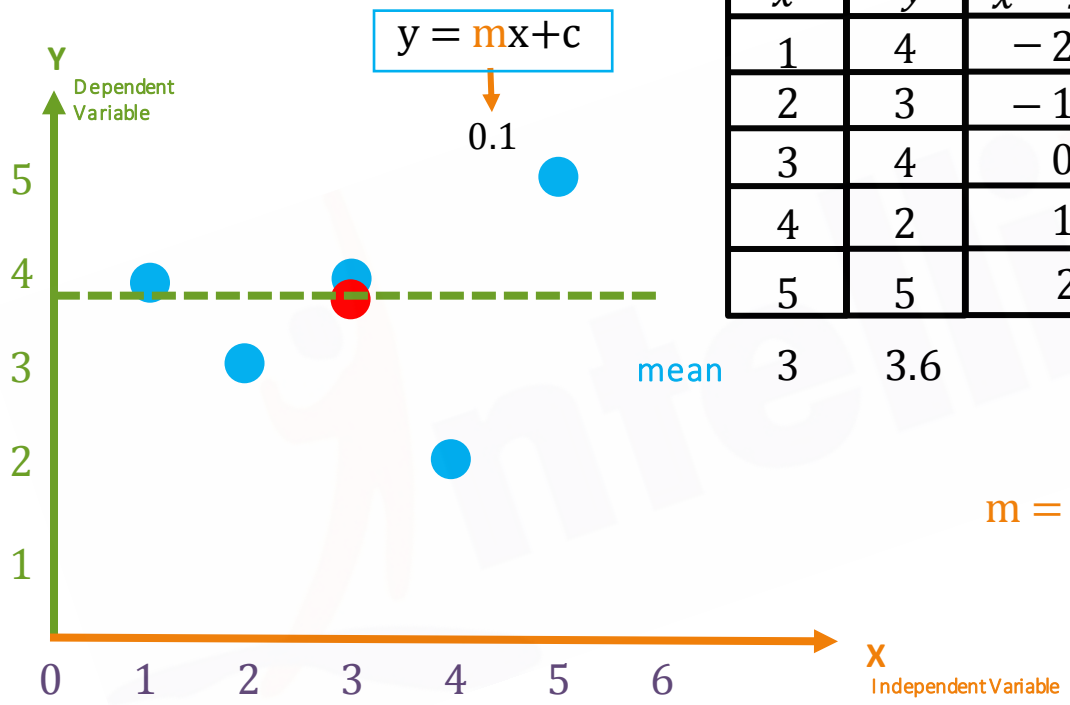
Understanding Linear Regression



x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean 3	3.6			$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

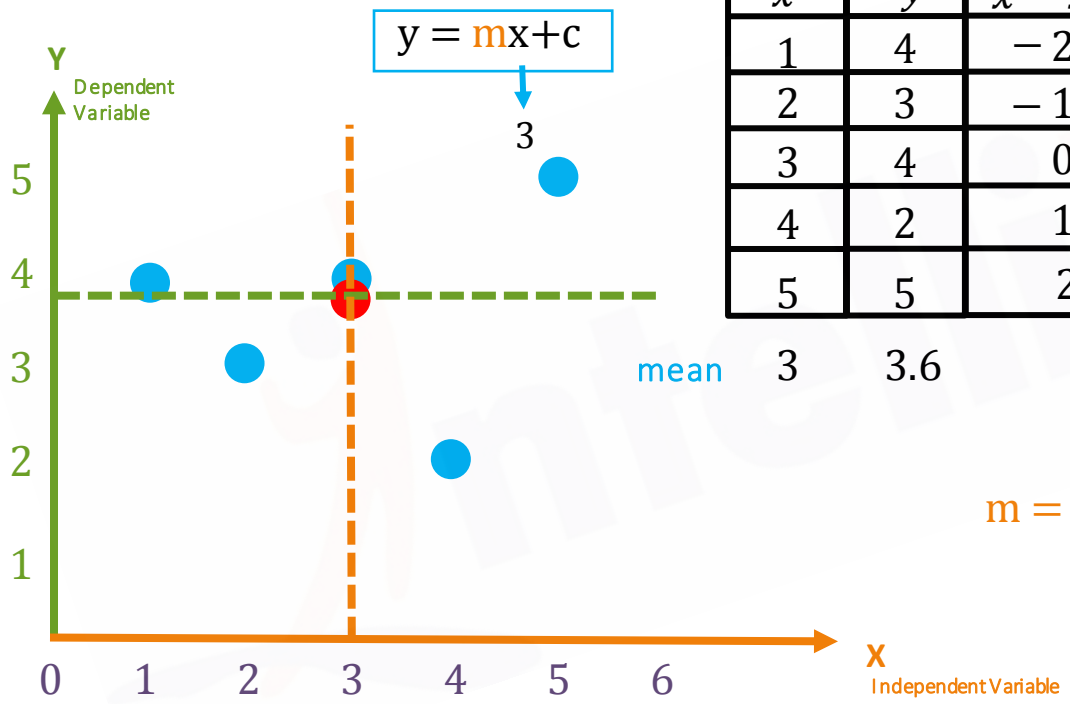
Understanding Linear Regression



x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression



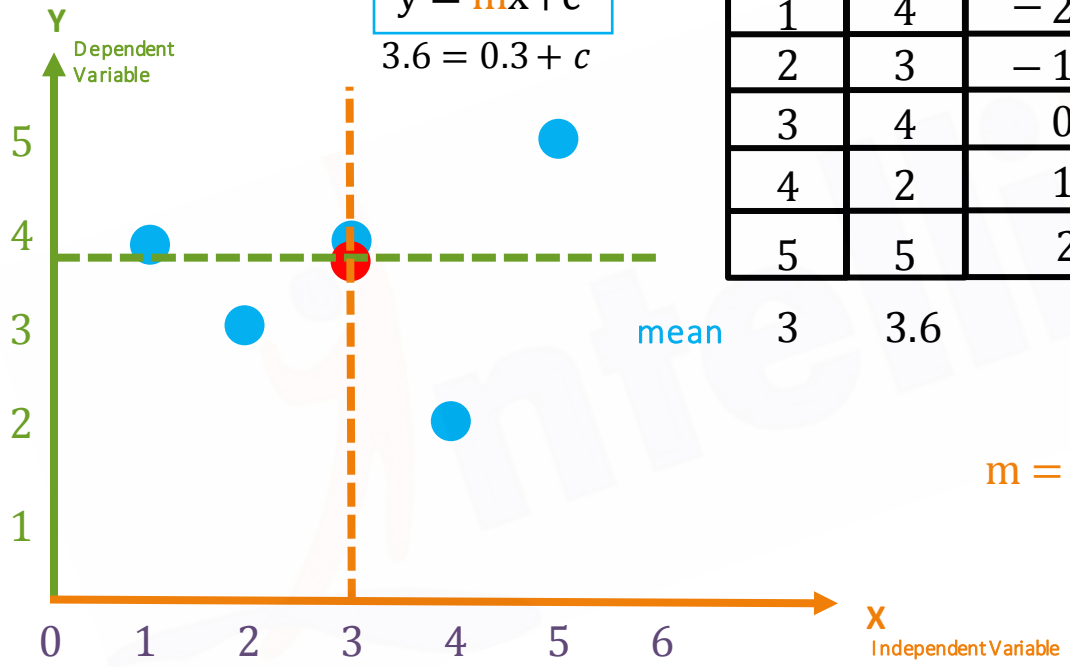
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression

$$y = mx + c$$

$$3.6 = 0.3 + c$$



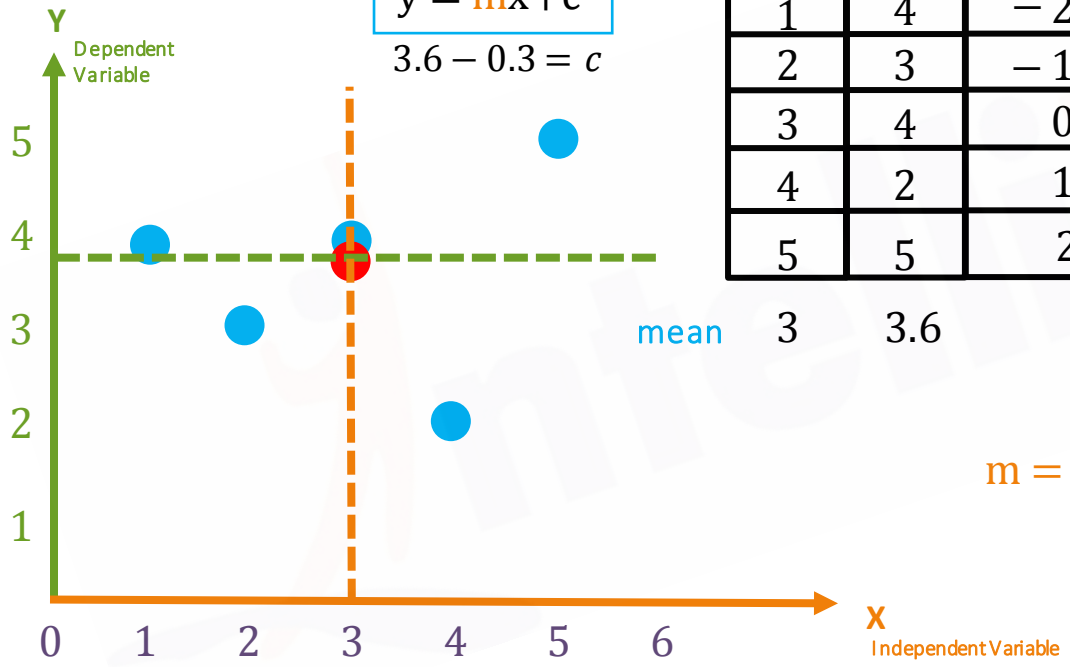
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression

$$y = mx + c$$

$$3.6 - 0.3 = c$$



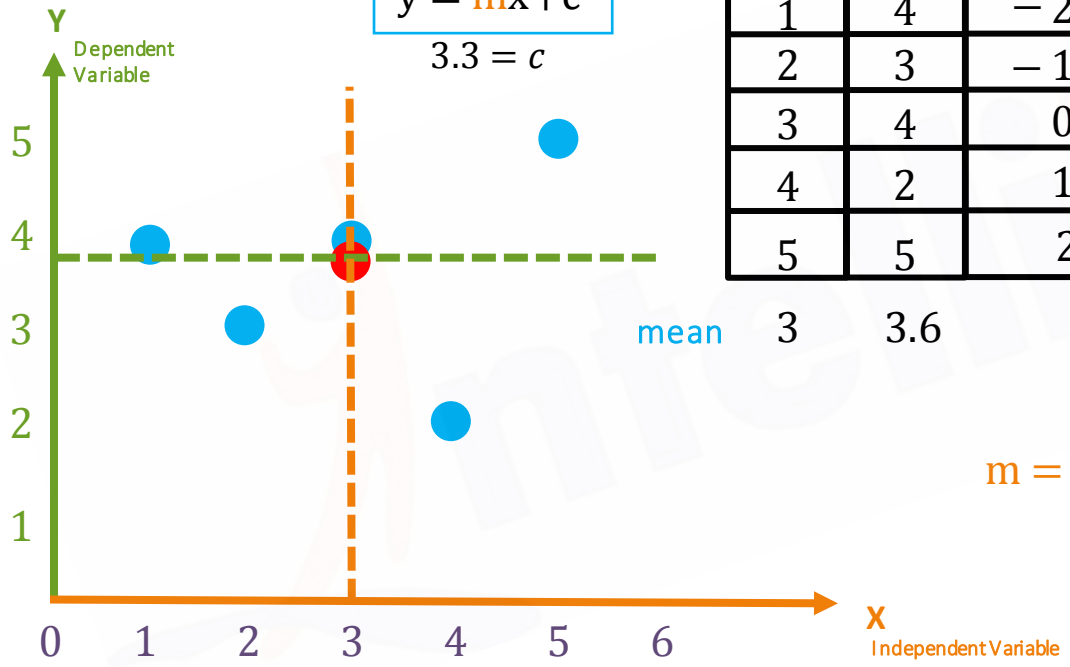
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression

$$y = mx + c$$

$$3.3 = c$$



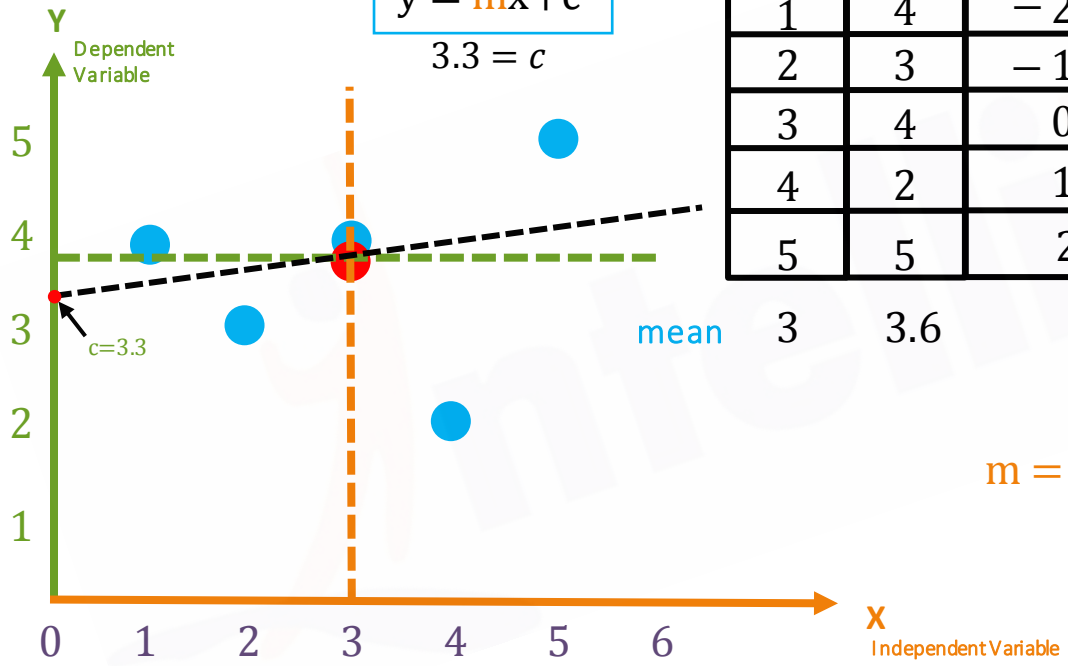
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression

$$y = mx + c$$

$$3.3 = c$$



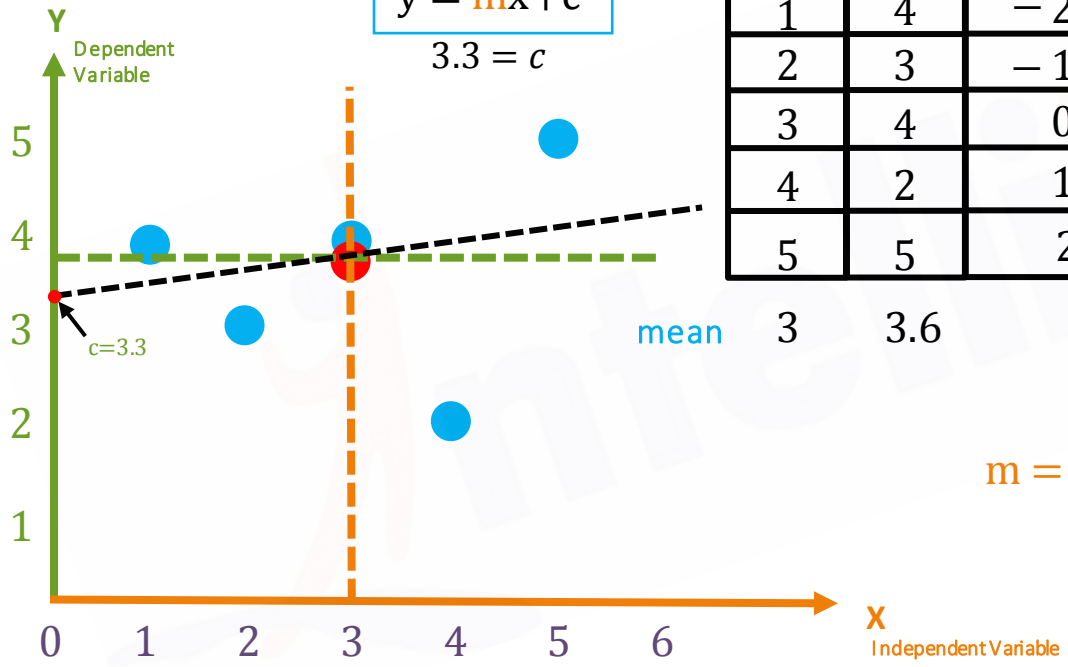
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2} = 0.1$$

Understanding Linear Regression

$$y = mx + c$$

$$3.3 = c$$



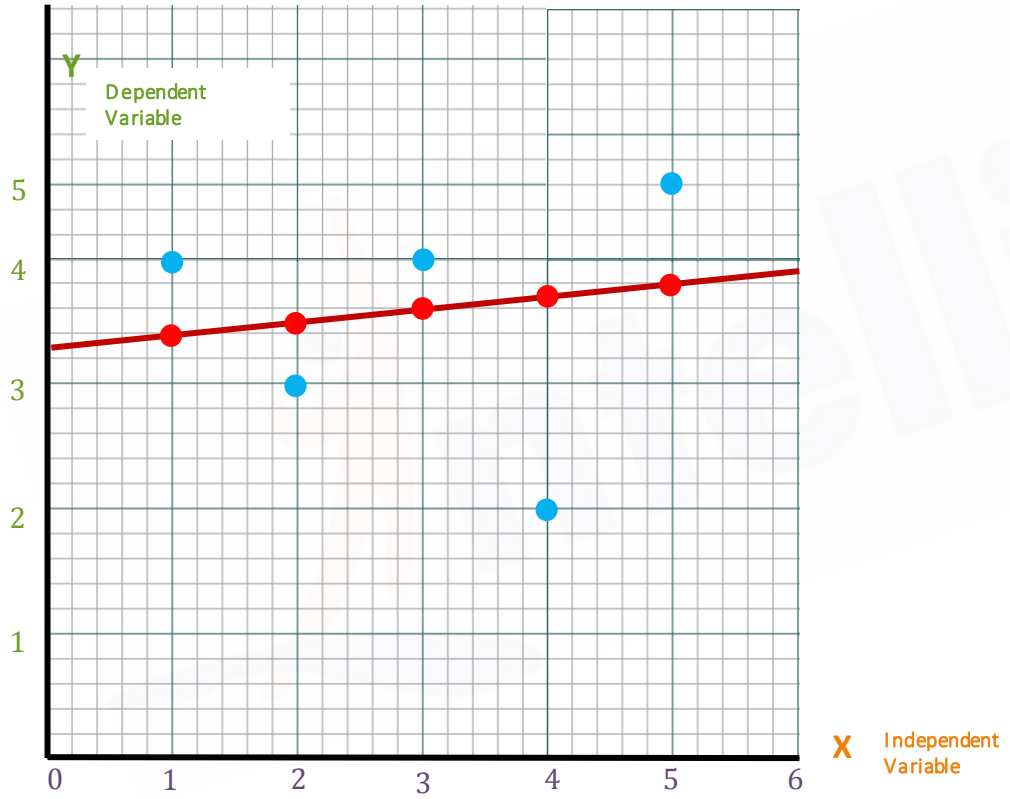
x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	4	-2	0.4	4	-0.8
2	3	-1	-0.6	1	0.6
3	4	0	0.4	0	0
4	2	1	-1.6	1	-1.6
5	5	2	1.4	4	2.8
mean	3	3.6		$\Sigma = 10$	$\Sigma = 1$

$$m = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2} = 0.1$$

Equation of Regression line for m = 0.1, c = 3.3 is:

$$y = 0.1x + 3.3$$

Mean Square Error

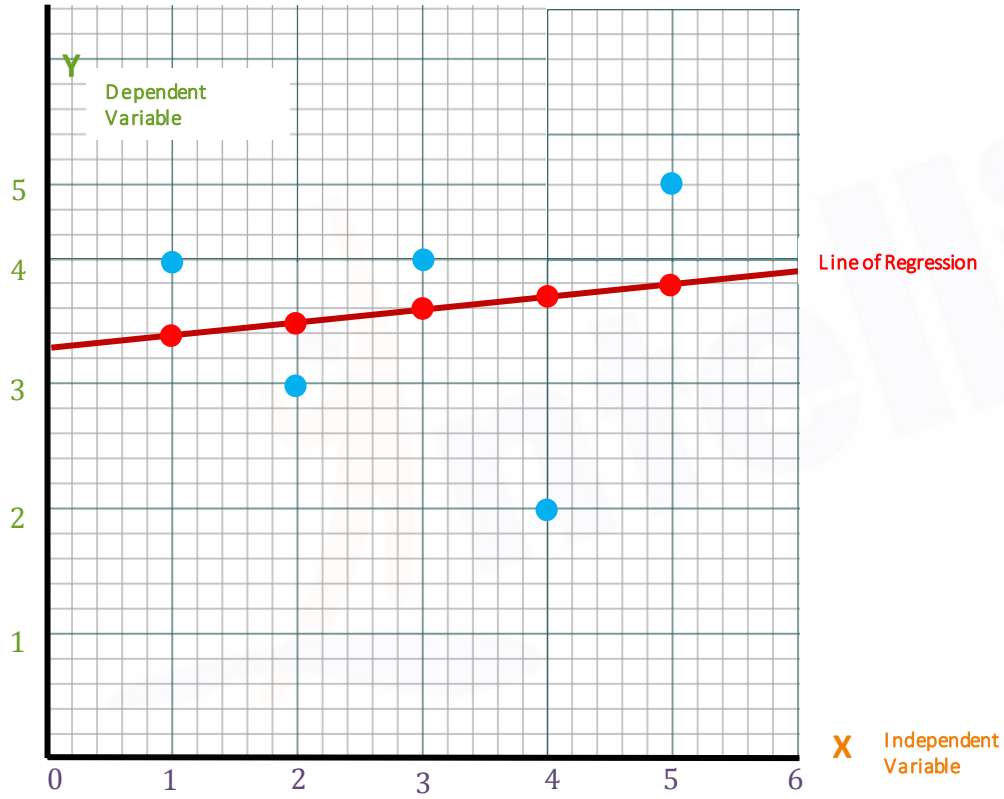


$$m = 0.1$$
$$c = 3.3$$
$$y = 0.1x + 3.3$$

For given $m = 0.1$ & $c = 3.3$,
Lets predict values for y when
$x = \{1,2,3,4,5\}$

$y = 0.1 \times 1 + 3.3 = \cancel{3.2}$ 3.4
$y = 0.1 \times 2 + 3.3 = \cancel{3.1}$ 3.5
$y = 0.1 \times 3 + 3.3 = \cancel{3.0}$ 3.6
$y = 0.1 \times 4 + 3.3 = \cancel{2.9}$ 3.7
$y = 0.1 \times 5 + 3.3 = \cancel{2.8}$ 3.8

Mean Square Error

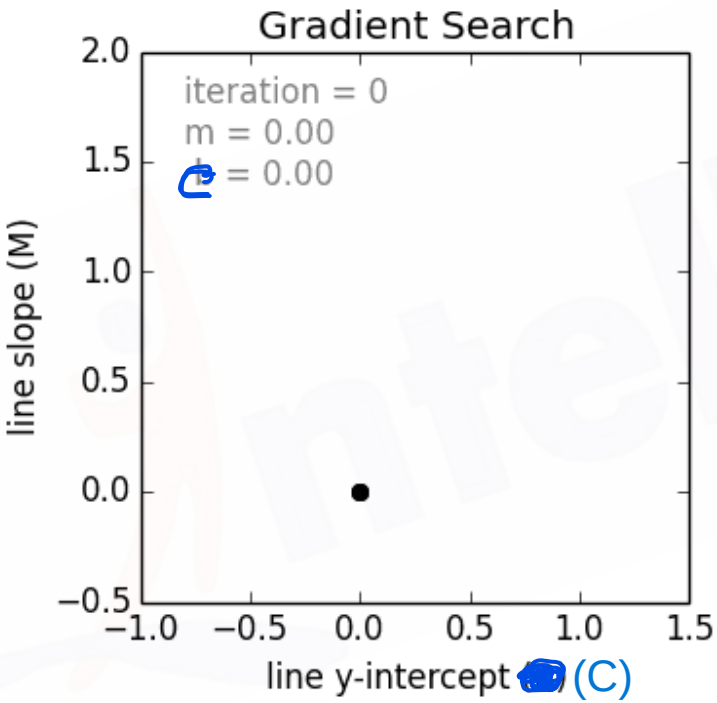


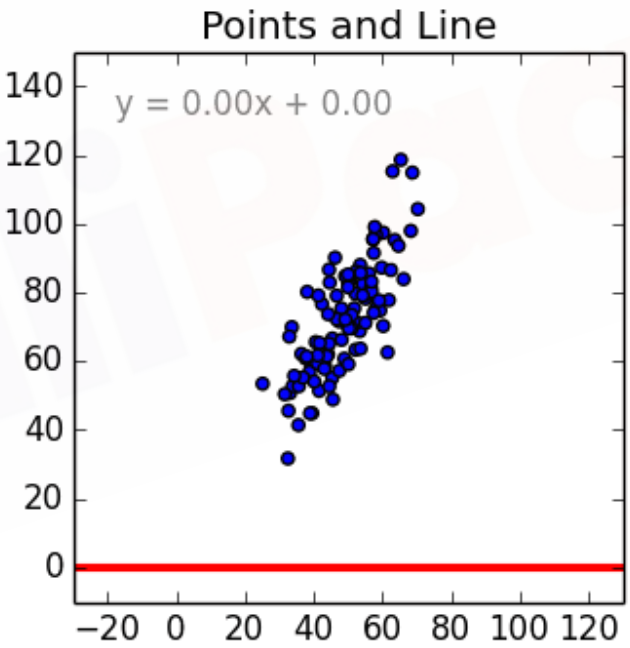
$$m = 0.1$$
$$c = 3.3$$
$$y = 0.1x + 3.3$$

For given $m = 0.1$ & $c = 3.3$,
Lets predict values for y when
$x = \{1,2,3,4,5\}$

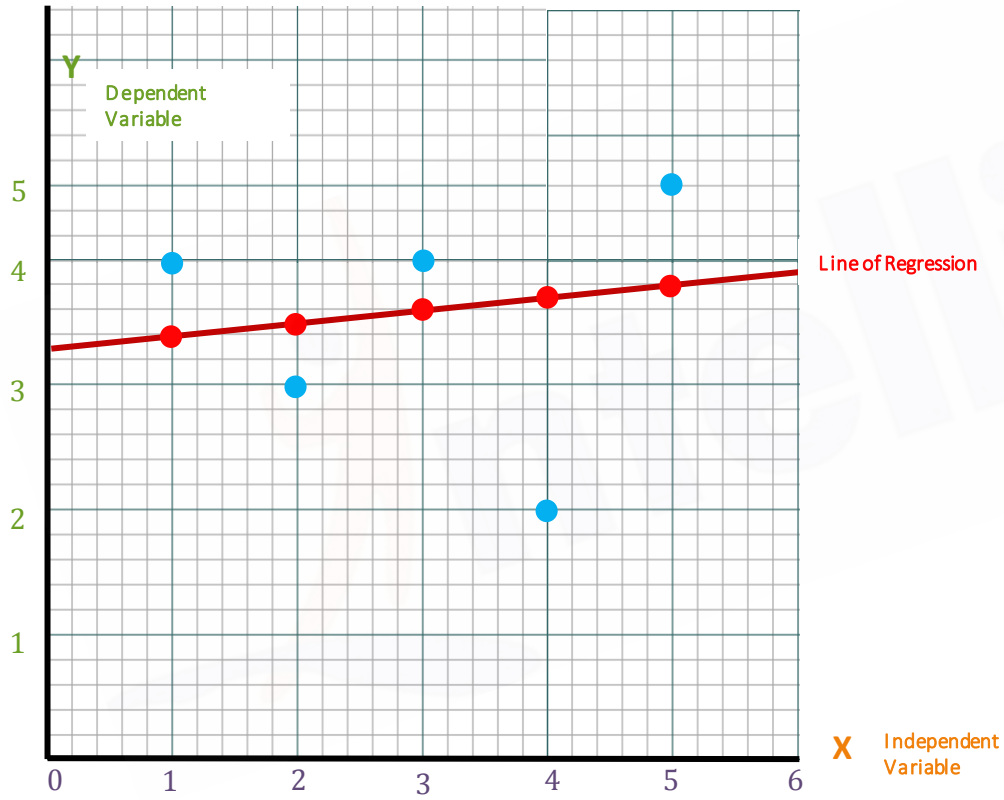
$y = 0.1 \times 1 + 3.3 = 3.2$ → 3.4
$y = 0.1 \times 2 + 3.3 = 3.1$ → 3.5
$y = 0.1 \times 3 + 3.3 = 3.0$ → 3.6
$y = 0.1 \times 4 + 3.3 = 2.9$ → 3.7
$y = 0.1 \times 5 + 3.3 = 2.8$ → 3.8

Finding the best Fit Line





Goodness of Fit – R^2

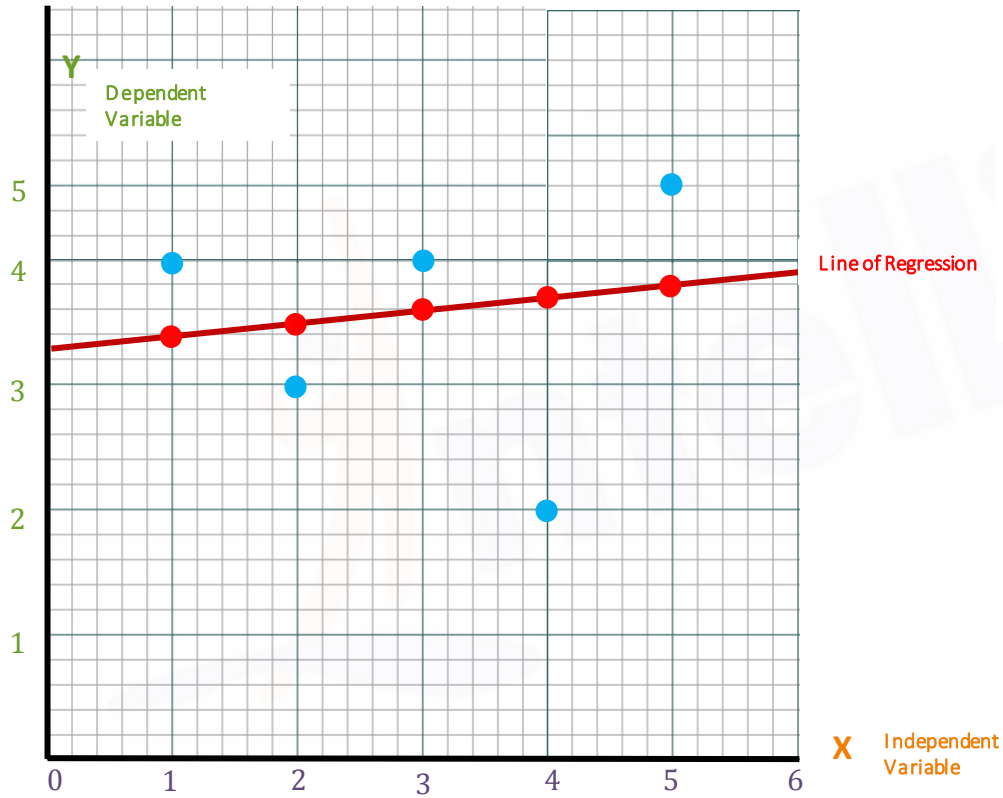


x	y_p
1	~~3.2~~ 3.4
2	~~3.1~~ 3.5
3	~~3.0~~ 3.6
4	~~2.9~~ 3.7
5	~~2.8~~ 3.8

$$R^2 = \frac{\sum(\text{Predicted Distance} - \text{Mean})^2}{\sum(\text{Actual Distance} - \text{Mean})^2}$$

$$R^2 = \frac{\sum(y_p - \bar{y})^2}{\sum(y - \bar{y})^2}$$

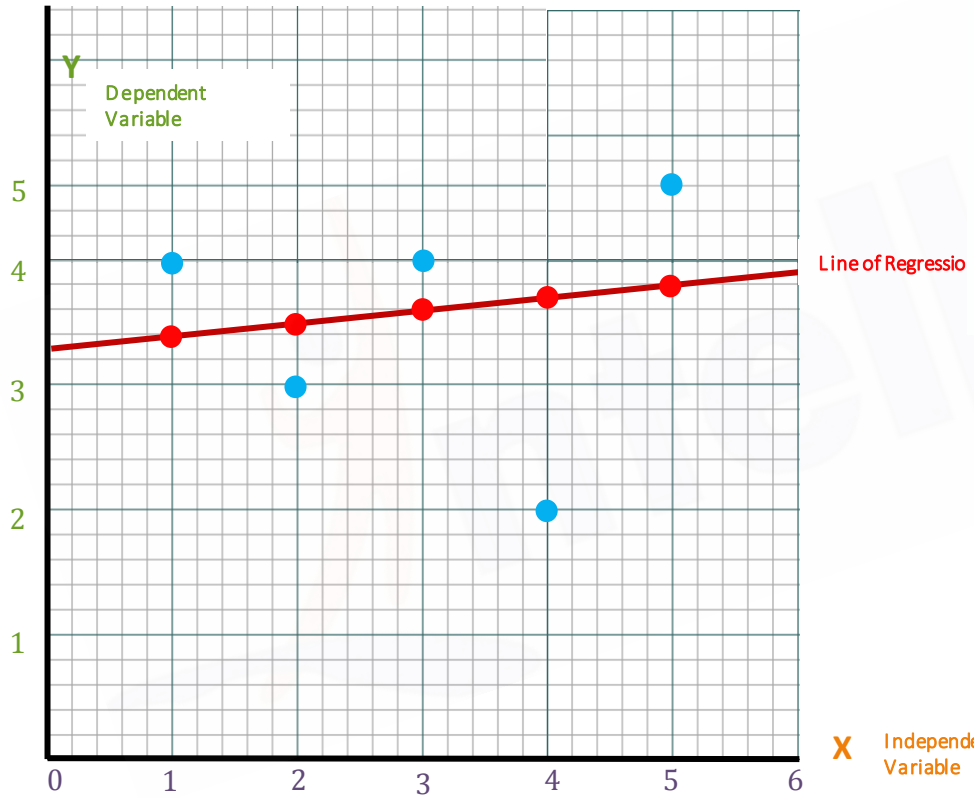
Goodness of Fit – R^2



x	y	y_p	$y_p - \bar{y}$
1	4	3.4	−0.2
2	3	3.5	−0.1
3	4	3.6	0.0
4	2	3.7	+0.1
5	5	3.8	+0.2

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

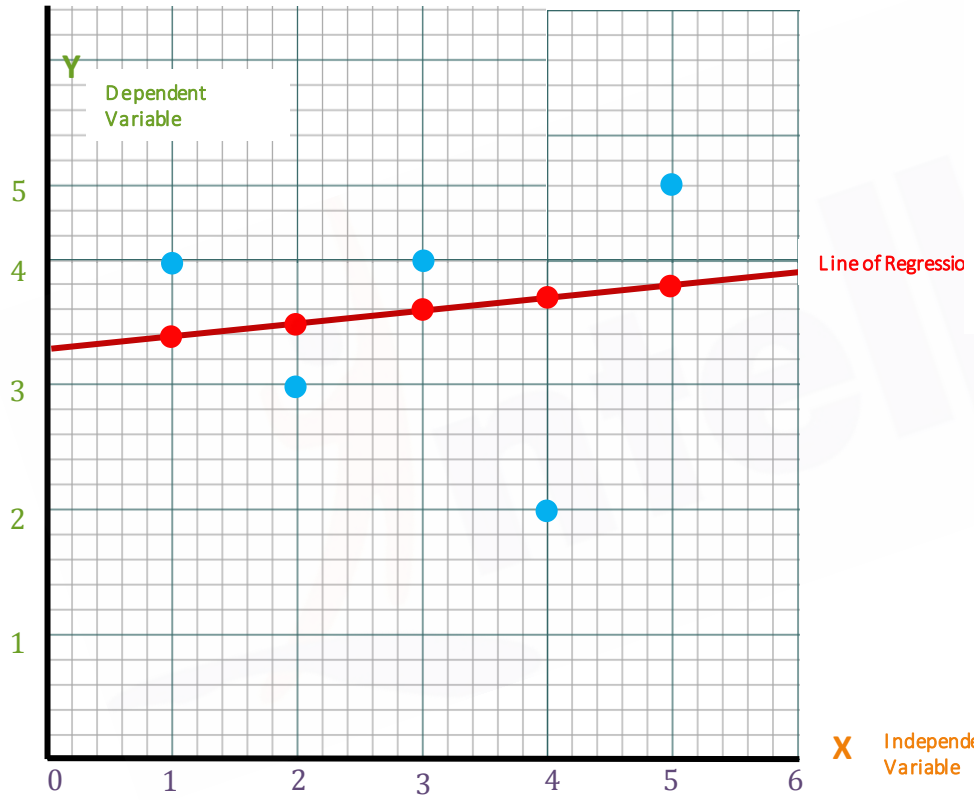
Goodness of Fit – R^2



x	y	y_p	$y_p - \bar{y}$	$y - \bar{y}$
1	4	3.4	−0.2	+0.4
2	3	3.5	−0.1	
3	4	3.6	0.0	
4	2	3.7	+0.1	
5	5	3.8	+0.2	

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

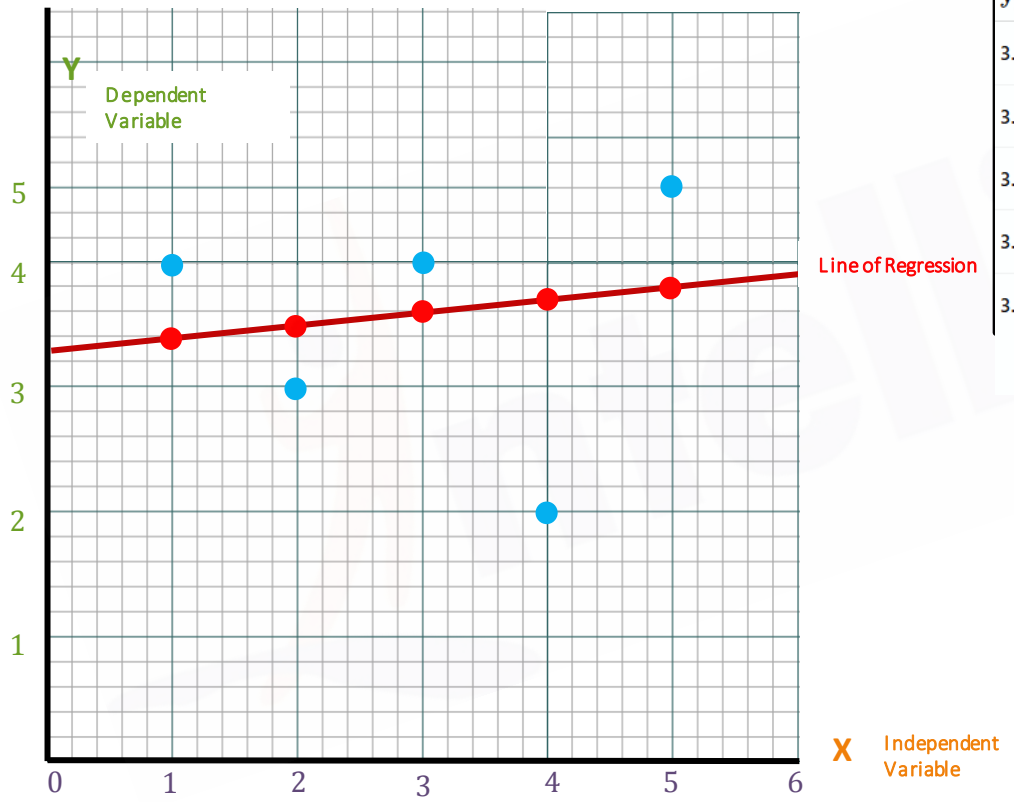
Goodness of Fit – R^2



x	y	y_p	$y_p - \bar{y}$	$y - \bar{y}$
1	4	3.4	−0.2	+0.4
2	3	3.5	−0.1	−0.6
3	4	3.6	0.0	+0.4
4	2	3.7	+0.1	−1.6
5	5	3.8	+0.2	+1.4

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

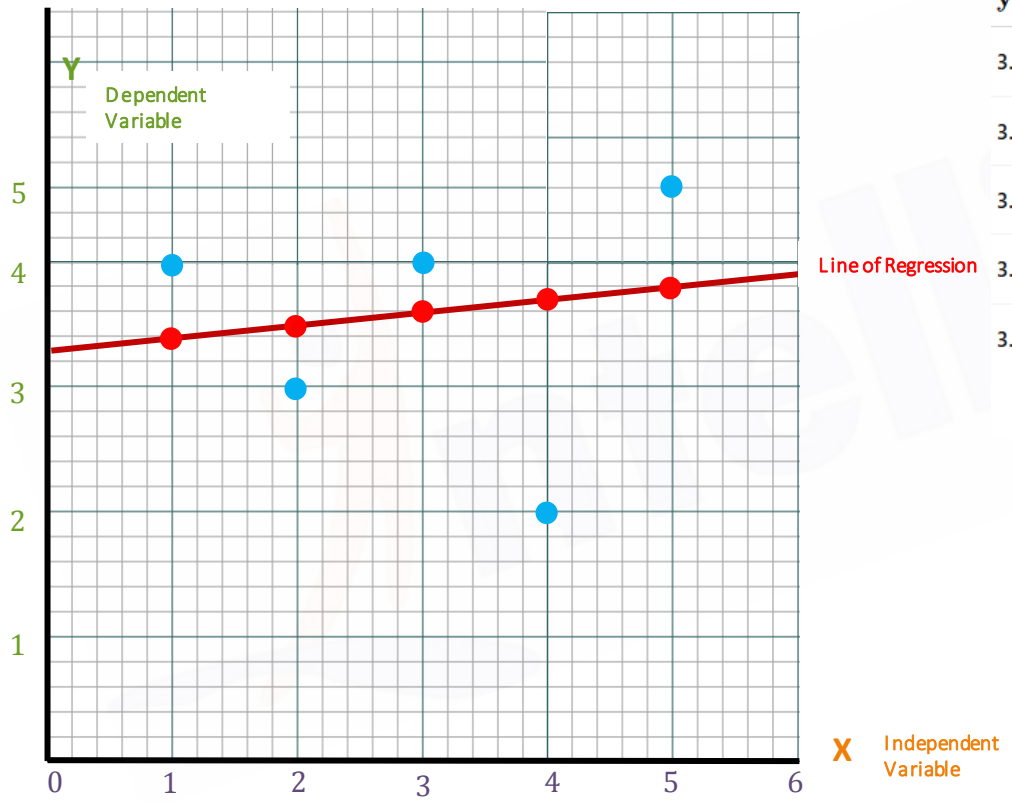
Goodness of Fit – R^2



y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$
3.4	−0.2	0.04	+0.4
3.5	−0.1	0.01	
3.6	0.0	0.00	
3.7	+0.1	0.01	
3.8	+0.2	0.04	

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

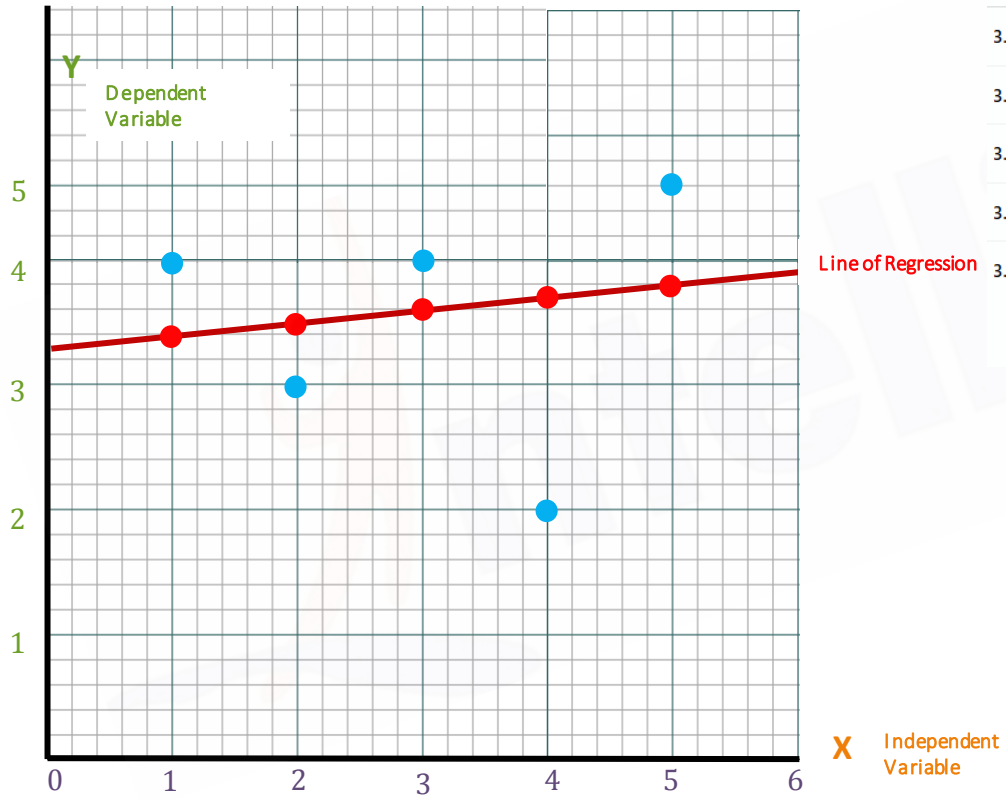
Goodness of Fit – R^2



y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$
3.4	−0.2	0.04	+0.4
3.5	−0.1	0.01	−0.6
3.6	0.0	0.00	+0.4
3.7	+0.1	0.01	−1.6
3.8	+0.2	0.04	+1.4

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

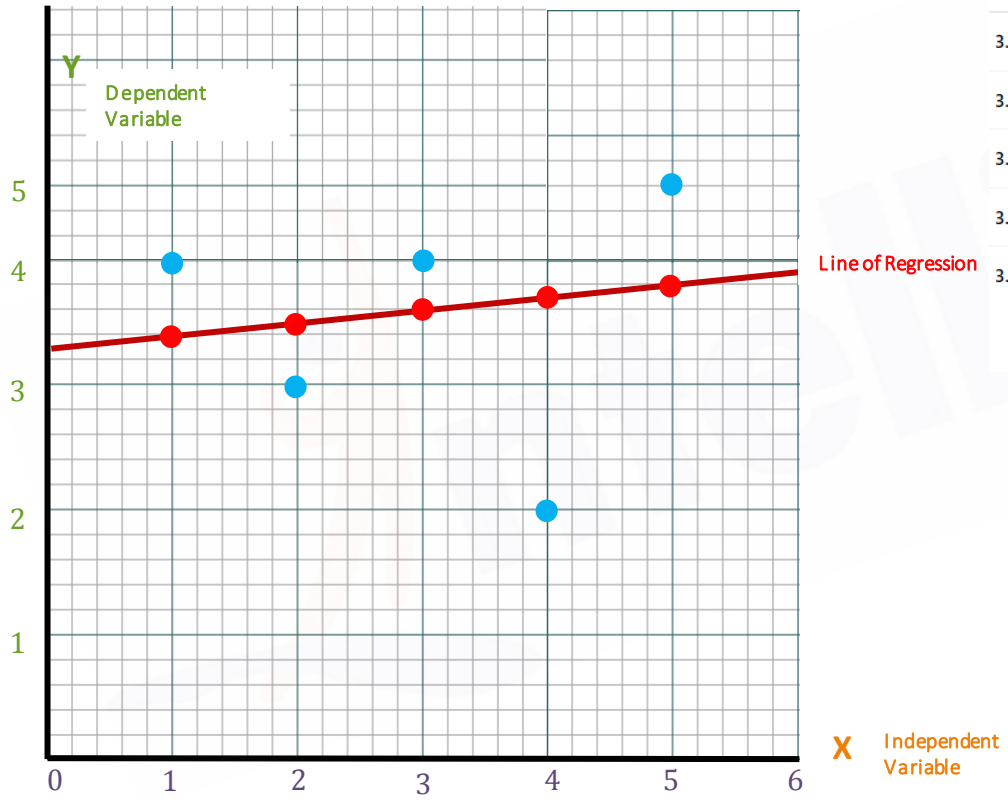
Goodness of Fit – R^2



y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$	$(y - \bar{y})^2$
3.4	−0.2	0.04	+0.4	0.16
3.5	−0.1	0.01	−0.6	
3.6	0.0	0.00	+0.4	
3.7	+0.1	0.01	−1.6	
3.8	+0.2	0.04	+1.4	

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

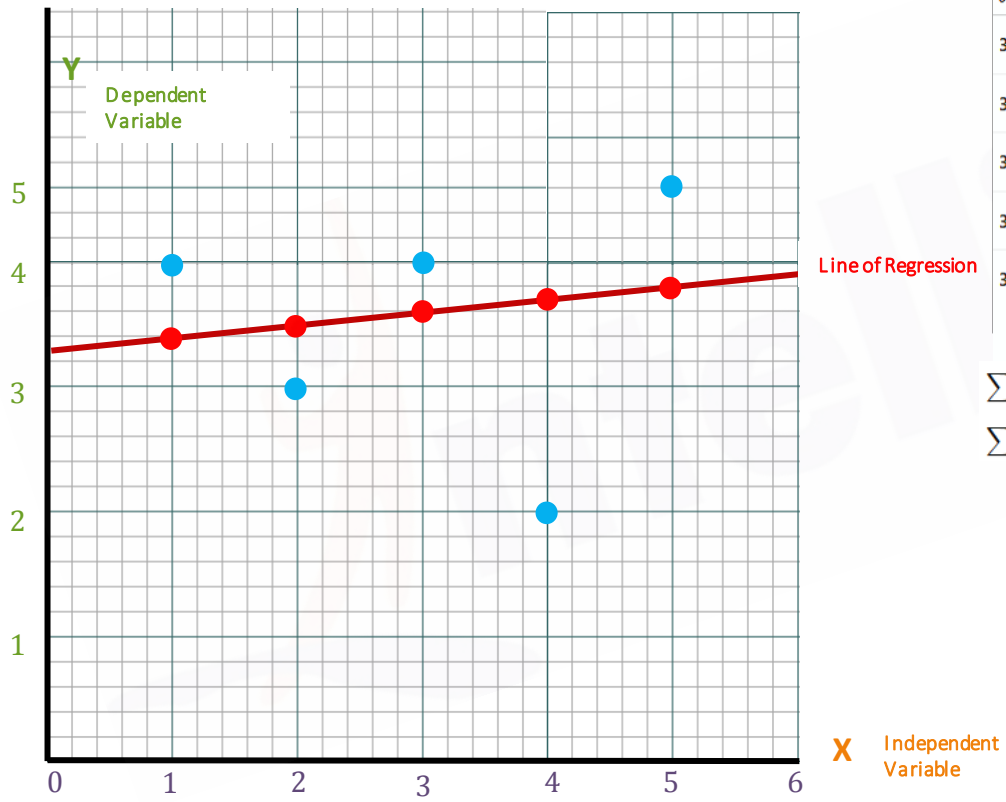
Goodness of Fit – R^2



y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$	$(y - \bar{y})^2$
3.4	−0.2	0.04	+0.4	0.16
3.5	−0.1	0.01	−0.6	0.36
3.6	0.0	0.00	+0.4	0.16
3.7	+0.1	0.01	−1.6	2.56
3.8	+0.2	0.04	+1.4	1.96

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

Goodness of Fit – R^2



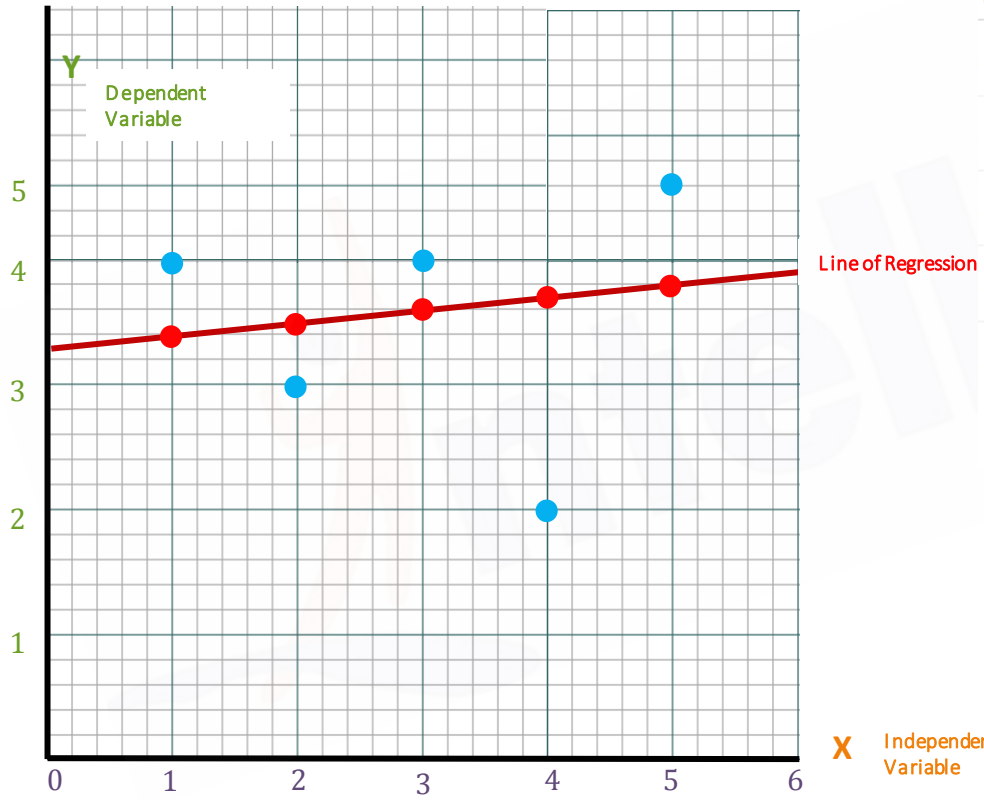
y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$	$(y - \bar{y})^2$
3.4	−0.2	0.04	+0.4	0.16
3.5	−0.1	0.01	−0.6	0.36
3.6	0.0	0.00	+0.4	0.16
3.7	+0.1	0.01	−1.6	2.56
3.8	+0.2	0.04	+1.4	1.96

$$\sum(y-\bar{y})^2 = 0.16 + 0.36 + 0.16 + 2.56 + 1.96 = 5.2$$

$$\sum(y_p-\bar{y})^2 = 0.04 + 0.01 + 0.00 + 0.01 + 0.04 = 0.10$$

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2} = \frac{\cancel{1.9} \rightarrow 0.10}{5.2}$$

Goodness of Fit – R^2



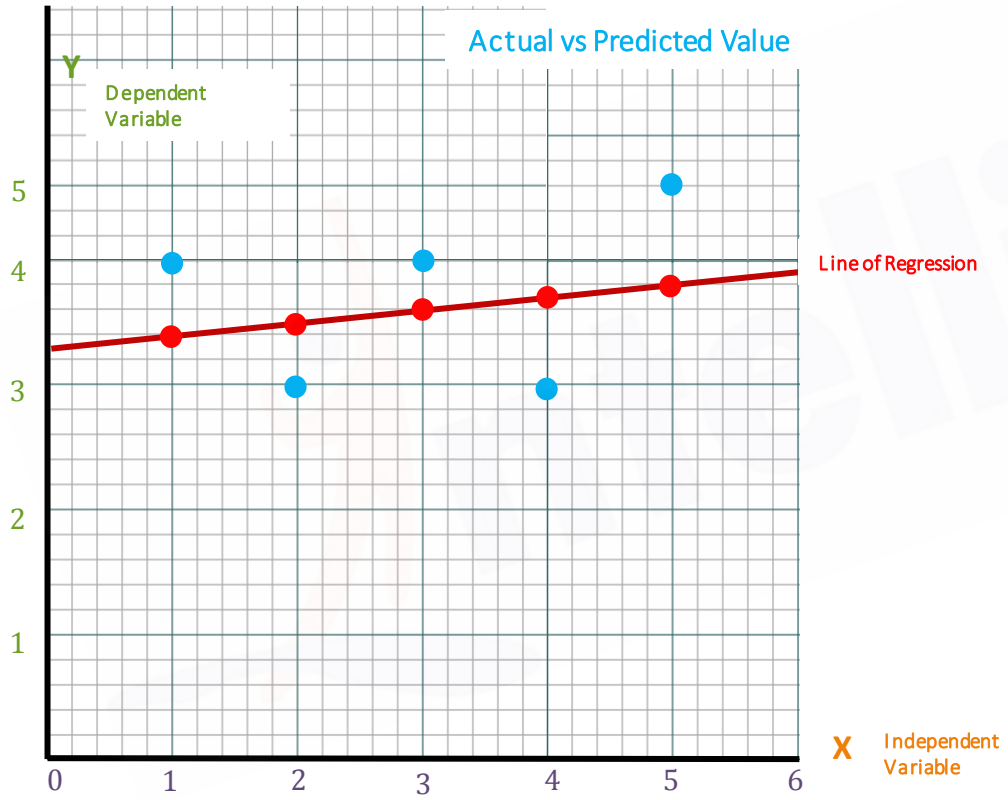
y_p	$y_p - \bar{y}$	$(y_p - \bar{y})^2$	$y - \bar{y}$
3.4	−0.2	0.04	+0.4
3.5	−0.1	0.01	−0.6
3.6	0.0	0.00	+0.4
3.7	+0.1	0.01	−1.6
3.8	+0.2	0.04	+1.4

$$\sum(y-\bar{y})^2 = 0.16 + 0.36 + 0.16 + 2.56 + 1.96 = 5.2$$

$$\sum(y_p-\bar{y})^2 = 0.04 + 0.01 + 0.00 + 0.01 + 0.04 = 0.10$$

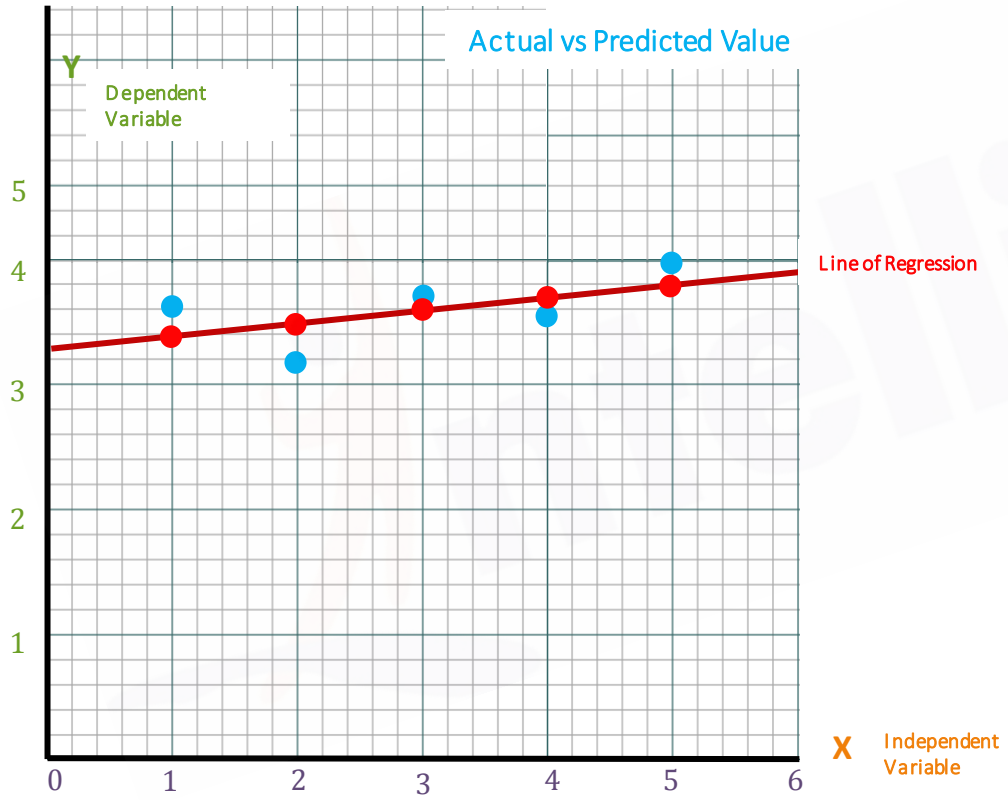
$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2} = \cancel{0.36} \rightarrow 0.019$$

Goodness of Fit – R^2



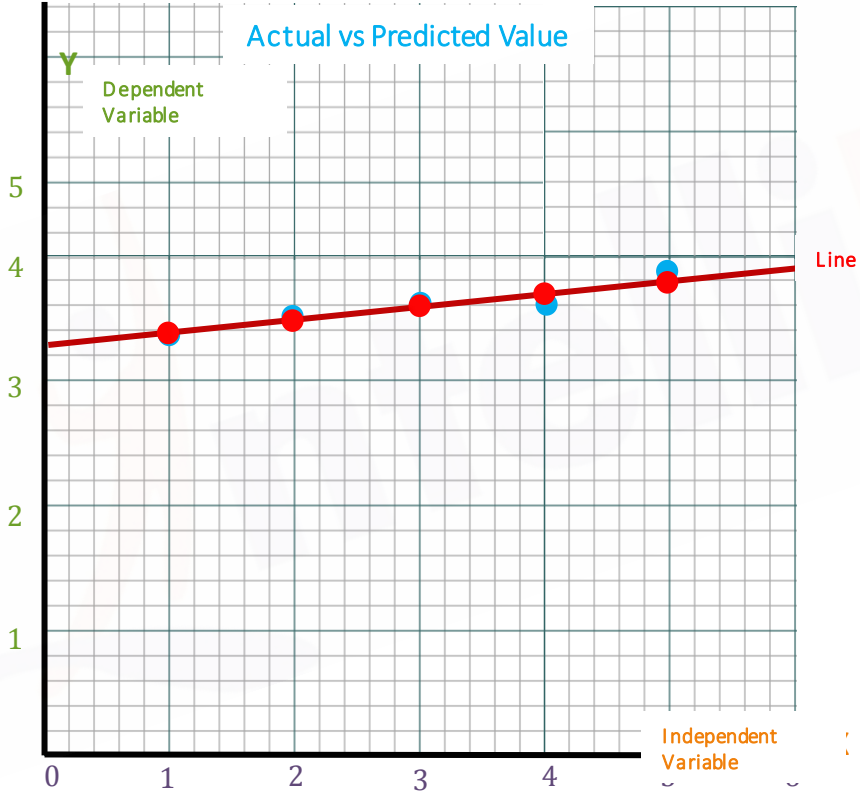
$$R^2 \approx 0.6$$

Goodness of Fit – R^2



$$R^2 \approx 0.9$$

Goodness of Fit – R^2



$R^2 \approx 1$

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